Home Interior Design & Office

Project ID: BSIT22-G29

BY

Atiqah Munir 076936

Javairia Shahzadi 076945

Areeba Nisar 076960

Under the Supervision of

Prof. Muhammad Asif

Bachelor of Science in IT(2022-2026)

Department of Computer Science and Information Technology

Govt. Islamia Graduated College Gujranwala

Home interior Design & Office

A project presented to

University of the Punjab, Lahore

In partial fulfillment

of the requirement for the degree of

Bachelor of Science in IT(2022-2026)

BY

Atiqah Munir	076936
Javairia Shahzadi	076945
Areeba Nisar	076960

Department of Computer Science and Information Technology,

Govt. Islamia Graduated College Gujranwala

DECLARATION

We hereby declare that this software, neither whole nor as a part has been copied out from any source. It is further declared that we have developed this software and accompanied report entirely on the basis of our personal efforts. If any part of this project is proved to be copied out from any source or found to be reproduction of some other, we will stand by consequences. No portion of the work presented has been submitted of any application for any other degree or qualification of this or any other university or institute of learning.

Signature:

Name:

Signature:

Name:

Certificate of approval

It is to certify that the final year design project (FYDP) of "Home interior design and office" was developed by **"Name"** and under the supervision of **"Prof Muhammad Asif"** in my opinion; it is fully adequate, in scope and quality for the degree of Bachelor of Science in IT.

Signature: ----------

FYDP SUPERVISIOR:

Signatures (Faculty Advisory committee (FAC)

Signatures			
Name			
	FAC1	FAC2	FAC3

Signature: ----------

Head of FYDP Coordination Office:

Signature: ---------- Dated: ----------

Chairperson, Department of [BSIT]

Executive Summary

Home interior design project have a basic reality that a user interior design digitally house or office means that inside room or office space make design in computer screen. For Example bedroom, drawing room , TV launch or Office area .Website provide user friendly interface also provide a login/signup option for user make a profile and then user start a design process . The user selects a different design style Modern, Traditional, Rustic, Minimalist and select luxury.

System provide functionality to user preview the design in 2D or 3D view the place where user select furniture and color theme according to wish. Website provide feature the user upload photos then system extract color palette and then suggest a furniture and wall color. Last that user saves or share design.

Acknowledgment

The acknowledgment section is an opportunity to express gratitude and appreciation to individuals and organization who have contributed to the completion of your Final year project (FYP).

Signature:

Name

Signature:

Name

Abbreviations

All the abbreviation and acronyms used in our FYP along their respective brief description.

Abbreviations	Full Form	Description
AI	Artificial intelligence	Enables the system to generate room design automatically based on user prompts
API	Application programming interface	Allow communication between frontend and AI services
CSS	Cascading Style Sheet	Used to style and design of web pages
DB	Database	Store user data, design preferences and generated result
HTML	Hypertext Markup language	Used to create a structure of web pages
HTTP	Hypertext Transfer Protocol	Protocol use for communication between client and server
JS	JavaScript	Adds interactive and dynamic behavior to the website
UI	User Interface	The visual Elements through which users interact with the system
URL	Uniform Resource Locator	The address use to access resources on the internet
3D	Three Dimensional	Represent Object height, width and depth for realistic visualization
PHP	Hypertext Preprocessor	A server side scripting language user for backend operation such as database interaction and form processing

Table of Contents

Chapter 1 Introductio 1
1. Introduction 1
1.1 Problem Statement 1
1.2 Problem Solution 1
1.3 Objective of the Proposed System 1
1.4 Scope 1
1.5 System Component 2
1.6 Related System Analysis/Literature Review 3
1.7 Vision Statement 3
1.9 Tools and Technologies 4
1.10 Project planning 5
Summary 6
2. Analysis 8
2.1 User class and characteristics 8
2.2 Requirement Identifying Technique 8
2.3 Functional requirement 11
2.3.1 Functional Requirement 12
2.4 Non-Functional Requirements 13
2.4.1 Reliability 14
2.4.2 Usability 14
2.4.3 Performance 14
2.4.4 Security 14
2.5 External Interface Requirements 15
2.5.1 User Interfaces Requirements 15
2.5.2 Software interfaces 15
2.5.3 Hardware interfaces 16
2.5.4 Communications interfaces 16
2.6 Summary 16
Chapter 3 Design and Architecture 17
3. System Design 18

3.1 Design consideration 18
3.2 Design Model 19
3.3 Architectural design 27
3.4 Data design 29
3.4.1 Data dictionary 31
3.5 User interface Design 34
3.5.1 Screen images 36
3.6 Behavioral Model 41
3.7 Design Decision 42
Summary 44
Chapter 4 Implementation 45
4. Implementation 46
4.1 Algorithm 46
4.2 External APIs/SDK 48
4.3 Code Repository 48
4.4 Summary 49
Chapter 5 Implementation 50
5. Introduction 51
5.1 Unit Testing (UT) 51
5.2 Functional Testing (FT) 52
5.3 Integration Testing (IT) 53
5.4 Performance Testing (PT) 54
5.5 Summary 56
Chapter 6 System Conversation 57
6. Introduction 58
6.1 Conversion method 58
6.2 Deployment 58
6.2.1 Data conversion 58
6.2.2 Training 59
6.3 Post deployment testing 59
6.4 Challenge 59
6.5 Summary 60
Chapter 7 Conclusion 61

7. Introduction 62
7.1 Evaluation 62
7.2 Traceability Matrix 62
7.3 Conclusion 63
7.4 Future Work 63
Appendix A Use Case Description 66
Use Case Template: AI interior design 66
Use Case Template: Design interior room 67
Appendix B General Coding standards & Guideline 70
1. Naming Convention 70
2. Scope Based Naming 70
3. Immutability 70
4. Controlled Access 70
5. Function Naming 70
6. Comments and Documentation 70
7. Code Indentation 70
8. File and Folder Naming 70
9. Class Structure 71
10. Code Organization 71
11. Scope Management 71
12. Access Modifiers 71
13. High Cohesion & low coupling 71
14. DRY Principal (Don't repeat yourself) 71
15. Single Responsibility Principal 71
16. Extensibility 71
17. Exception Handling 71
18. Logging 71
19. Consistent logging standards 71
Appendix C: Application Prototype 73
1. Prototype overview 73
2. Tools Used for Prototype 73
3. Screenshot of key interfaces 73
4. Navigation Flow 76

5. Responsive Design.....76

List of Figures

Figure 1 Use Case 1 10
Figure 2 Use Case 2 11
Figure 3 class diagram 20
Figure 4 Sequence Diagram UC 1 21
Figure 5 Sequence Diagram UC 2 22
Figure 6 Sequence Diagram UC 3 23
Figure 7 Sequence Diagram UC 4 24
Figure 8 Sequence Diagram UC 5 25
Figure 9 ER Diagram 26
Figure 10 Class Relationship Diagrams 27
Figure 11 UML Component Diagram 28
Figure 12 MVC diagram 29
Figure 13 State Transition Diagram 41

List of Tables

Table 1 Related System Analysis and Proposed Project Solution 3
Table 2 Tools and technologies for proposed project 4
Table 3 User classes and characteristics for proposed project 8
Table 4 user classes and characteristics for proposed project 8
Table 5 admin classes and characteristics for proposed project 9
Table 6 Description of FR-1 13
Table 7 Examples of Algorithm 46
Table 8 Details of API used in project 48
Table 9 unit testing 51
Table 10 functional testing 53
Table 11 Integration testing 54
Table 12 performance testing 56
Table 13 challenge 59
Table 14 Evaluation 62
Table 15 Traceability Matrix 62
Table 16 Conclusion 63
Table 17 Use Case 1 66
Table 18 Use Case 2 67



Chapter 1 Introduction

1. Introduction

The home interior design and office system is a web based application to help users design and visualize their home interior easily. Interior design requires professional help, time and cost. This system provides a digital platform where user can design their own choice or system generate the design through prompt according to user requirement. The user can select room style, room type, arrange furniture and visualize room in 3D.it make the design process simple, interactive and accessible for everyone.

1.1 Problem Statement

Many people face difficulty to designing their home interior due to lack of knowledge. The professional designer demand the high cost. Existing method are time consuming and do not provide real time visualization.

1.2 Problem Solution

There are some solutions to solve above problem:

- User provide facility to design virtual room
- Try the different layout
- Provide facility to customize the color and furniture
- Provide 3D visualization for better understanding
- System save the time and cost

1.3 Objective of the Proposed System

- Allow user to select to design multiple rooms in their home
- Allow user to choose different design styles(e.g. modern, classic)
- Provide a furniture catalog with beds, sofas, chairs, lamps
- Display design preview through 3D visualization
- Provide user with option to upload their room images
- Enable drag and drop functionality for easy placement of furniture
- Support saving and share user designs for future use

1.4 Scope

- The project covers complete home design including (bedroom, living room, kitchen and office)
- Predefine furniture items like beds, sofas, tables, lamps, chairs, and wardrobe will be available
- A suggested color palette will be included
- Users can drag-and-drop furniture items to arrange them as they want
- Ensure real-time visualization of room

- Also save and share final design
- AI-based automatic generate

Priorities:

- Easy design creation
- 3D visualization
- User customize feature

Stakeholders:

- Homeowner/Users: Customize their home which is flexible or cost effective.
- Admin: Responsible for system security, manage furniture and style data, control user and save their design.
- Furniture provide: wants their furniture collection to be displayed user and product visibility through the platform.
- Project Team: Create user-friendly, modern systems that improve the interior design experience.

Milestones

- Requirement analysis
- System design
- Implementation
- Testing
- Deployment

1.5 System Component

Module system /components of the project

These are feature of all modules are as follows:

1. **Client Web App(User Side)**
    - User registration and login module
    - Home dashboard module
    - Design style selection module
    - Room selection module
    - Furniture arrangement module
    - Save and share module
    - Upload image module

2. **Admin web app**
    - Admin login module
    - User management module (view/delete users)
    - Furniture management module (Add/delete/Update furniture items)
    - Design style management module
    - System monitoring module

3. **Furniture provider module**
    - Furniture list management
    - Product upload module (images, detail, categories)
    - Visibility tracking module

4. **Core System Services**
    - Authentication and security system
    - Database management system
    - 3D rendering Engine for visualization
    - Save and load design system

1.6 Related System Analysis/Literature Review

Here is the analysis of related system is as follows:

Table 1 Related System Analysis and Proposed Project Solution

Application Name	**Weakness**	**Proposed Project Solution**
Sketch UP	Require professional Skills and is complex for beginner	The proposed system provide a user friendly interface so even non-technical users can design interior easily
Room Planner/Planner 5D	Some feature are paid and customize is limited in free versions	The proposed system free access with essential design furniture like furniture placement and room customization
IKEA Home Planner	Limited furniture option and no advance real time 3D flexibility	The proposed system provide a wide range of furniture styles and better interactive 3D visualization

1.7 Vision Statement

Here a vision Statement for Home interior design and office

For [Homeowner and interior designer **]**

Who [want an easy and cost effective , efficient way to design and visualize interior space **]**

The [Home interior design and office **]**

Is [a web based interior design system**]**

That [allow user to select style, select room, arrange furniture and visualize in 3D **]**

Unlike [traditional design method and complex interior design system **]**

Our Products [provide a simple user friendly, and affordable solution that enables anyone design their dream space **]**

1.8 System Limitation and Constraint

Limitation:

- This plate will not work without internet
- 3D may be slow on low-end mobiles or computer
- Colors may look different on every screen so may be exact color did not match perfect

Constraint:

Scope: the scope of the system is to develop a secure reliable and user-friendly home interior design and offices user easily design our room's kitchen and offices etc. on choice and also design the home with the help of AI.

Time: every project has a deadline. This project design must be completed within a given time.

Client's requirements: the design must match the client's needs, preference and life style.

- Colors and theme
- Furniture style
- Layout decisions

1.9 Tools and Technologies

Mention all the hardware/software tools and technologies with version information used in implementation of the project. Write about the APIs, languages, SDK etc. which we will use for implementation.

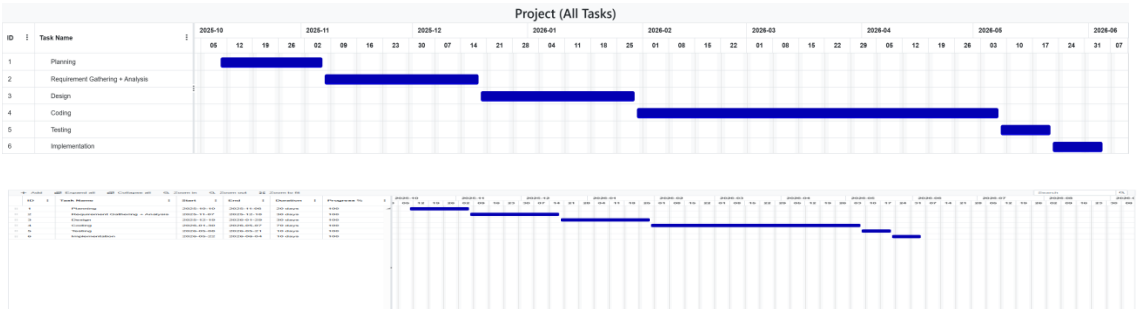
Table 2 Tools and technologies for proposed project

	Tools	Version	Rational
	Visual studio	VS Code :1.67.2	Writing and editing code download extension easily
	XAMPP	Latest	Local server setup for PHP+MySQL
	Github	Latest	To store project

Tools and Technologies			securely and track version
	Technologies	**Version**	**Rational**
	PHP	PHP-v 8.2.12	Server side scripting to handle user data processing and backend
	MySQL	MySQL-v8.4	To store user account, room selection, furniture data and saved design
	HTML5	---	Create structure of a web page
	CSS3	---	For designing and styling web pages
	Javascript	ES6+	Interactive drag and drop feature

1.10 Project planning

Gantt chart representing the plan of carry out the project activities, representing the milestone as well as resources require for each task:



Summary

Here's a brief summary of the Home Interior Design project:

Interior design focuses on making the interior space of place functional and beautiful. User can design multiple rooms such bedroom, kitchen, offices and living rooms while selecting furniture and color palettes is according to their preferences. The objective is to provide an interactive and user-friendly home interior design system where non-professional user plan easily room layout and decoration using drag-and-drop feature. The system will developed Using HTML, CSS, JavaScript & 3D visualization, while PHP and MySQL will be used for backend development and user authentication. User can also upload photo and system also recommend color palates It will also allow user to save and share design for future use.



Chapter 2 Requirement Analysis

2. Analysis

It outlines the systems objectives, scope, and the rationale behind the technology selection.

2.1 User class and characteristics

These are the user classes and their characteristics are as follow:

Table 3 User classes and characteristics for proposed project

User class	Characteristics
Client	Register and log in to the system. Non-technical user basic knowledge focuses on design.
Interior designer	Create and manage the design. Skilled in design technical knowledge and also creativity thinking
Admin	Manage the user accounts. Monitor system performance. Accesses to System logs and also responsible for system maintenance.

2.2 Requirement Identifying Technique

Each use case focuses on a specific goal or action the user can perform in the system.

Table 4 user classes and characteristics for proposed project

Use case Name	Associated Functional Requirement
Register user	Allow the client to create an account.
Login to the system	Enable users to log in the system securely.
Select room type	User select room type (home/office bedroom, kitchen, etc.)
Enter preferences	User provide preferences like color, style, furniture type etc.
Upload photo	User upload photo of room for matching color or furniture type.

Generate AI design	System generate interior design using AI based on user input other than user own choice.
Save and share	User can save final design and also share the design

Table 5 admin classes and characteristics for proposed project

Use case name	**Associated Functional Requirement**
Admin login	Admin can securely log into system.
Manage user	Admin can view, add, update and delete user account.
Add design	Admin can upload new interior designs for AI references.
Manage categories	Admin can create and manage categories (home/offices, bedroom, kitchen, etc.)
System monitoring	Admin can monitor system activities and usage
View feedback	Admin can view user feedback and suggestions.

Use Case Diagram: AI interior design

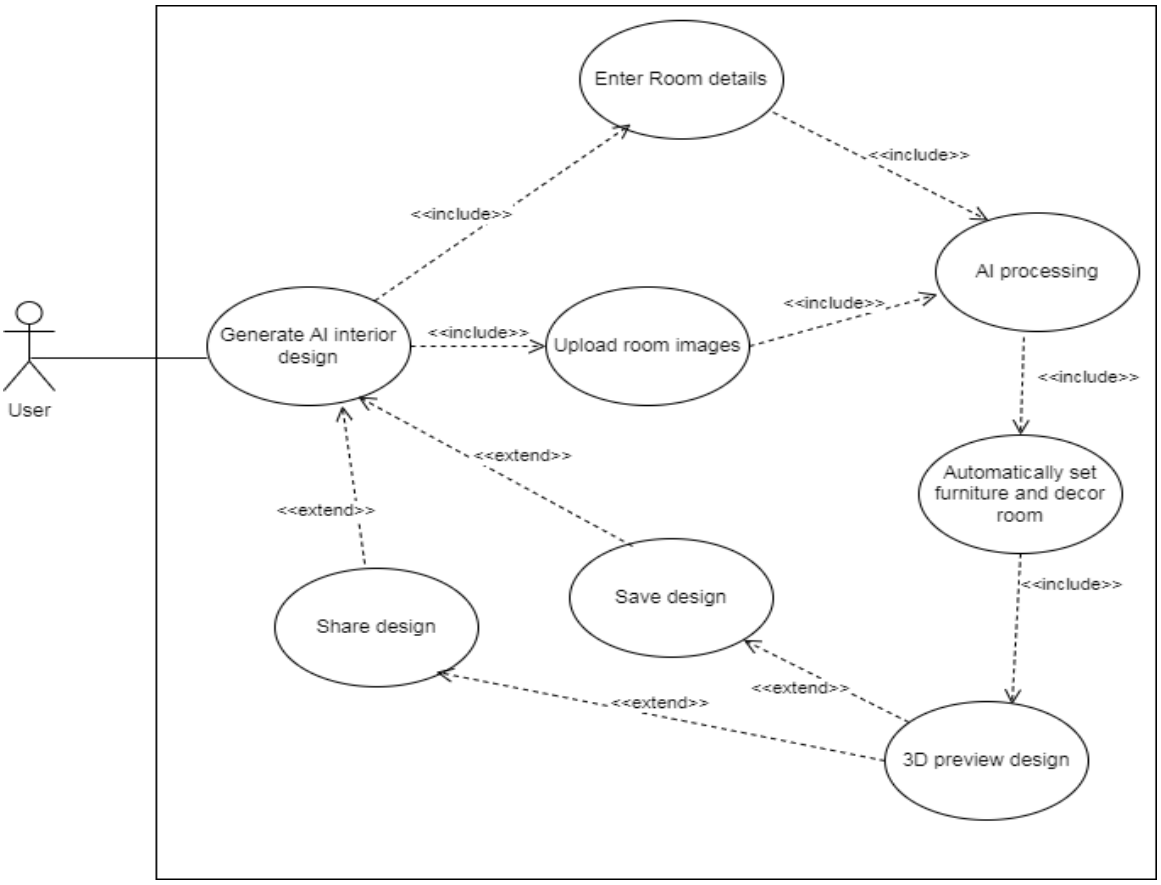


Figure 1 Use Case 1

Use Case Diagram: **Design interior room**

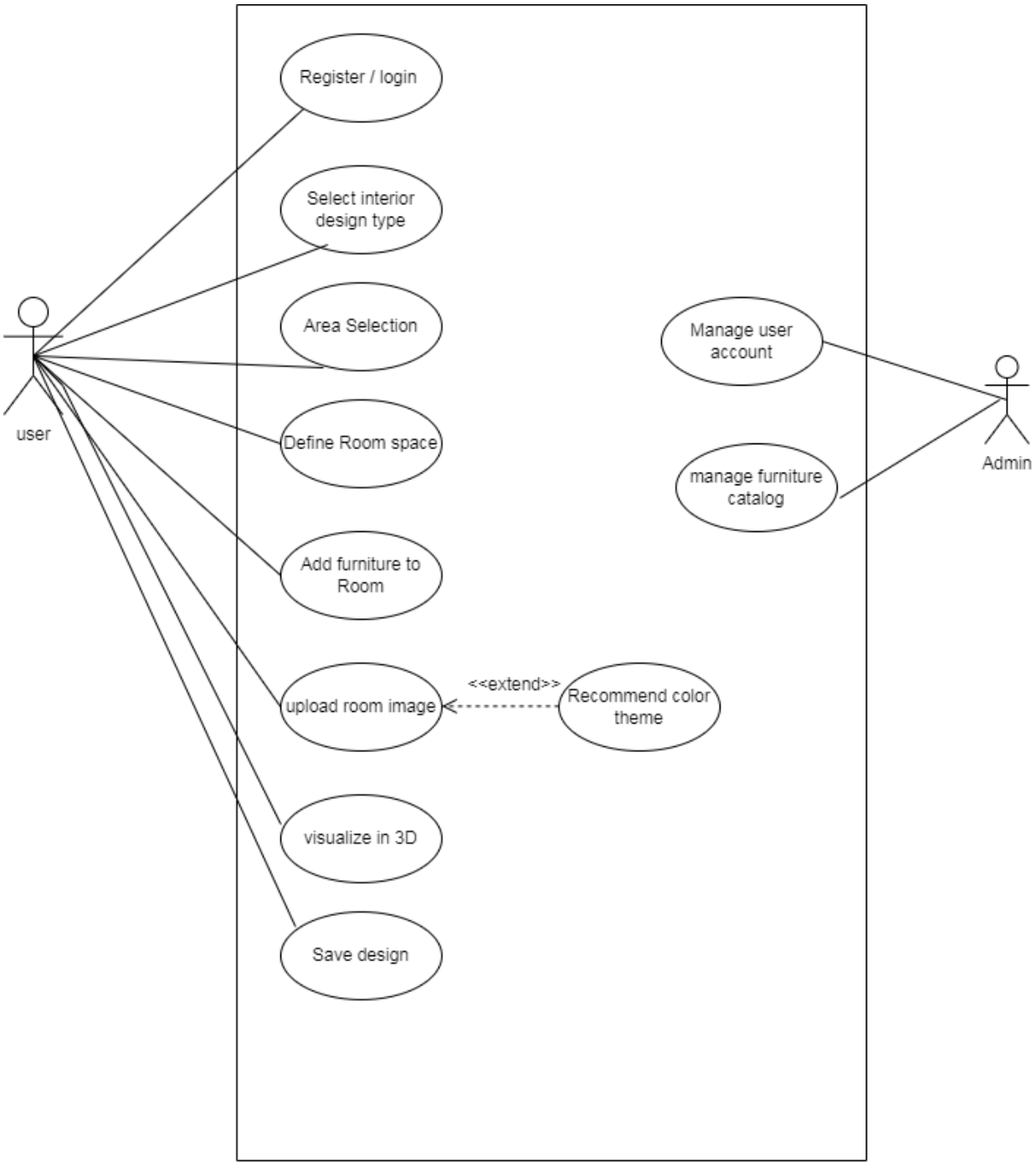


Figure 2 Use Case 2

2.3 Functional requirement

The following are the core **functional requirements** of the home interior design and office. These describe how the system should behave in response to user action.

1. **User registration and login**
    - The system will allow user to register and create account.
    - System will allow users to log in and log out securely.

2 Select design

- System will allow user to select room type (home/office bedroom, kitchen, etc.)
- System will user to enter design preferences such as color, style and furniture type.

3 Upload photo

- System will allow to the user to upload photo.
- System will analyze the photo to suggested matching furniture and color combination.
- System will use upload images to improve AI design accuracy.

4 AI based design

- System will generate interior designs using AI based on user input and uploaded images.
- System will provide multiple design options to the user.

5 3D view

- System will allow to user to see final design in 3D view.
- System will allow the user drag and drop the furniture.

6 Save and share design

- System will allow users to save their design.
- System will allow the user share the save design.

7 Admin panel

- System will allow admin to manage user accounts (add, update, delete).
- System will allow admin to manage designs.
- System will allow admin to manage design categories.
- System will allow admin to manage AI dataset and training

2.3.1 Functional Requirement

Here's the functional requirement specification for a home interior design and office design system.

1. **User registration and login** user registration and login name and password for create an account.
2. **Admin login** admin can log in and manage system data.
3. **Create design** the system will allow user to can design rooms by selecting type and layout.
4. **Upload image** user can upload room image foe design help.
5. **AI suggestion** system will provide AI based design ideas.

6. **View design** user can view be view design in 2D/3D.
7. **Save and share design** the system will allow the user save and share the final design

Table 6 Description of FR-1

Identifier	FR-1
Title	User registration and login
Requirements	The system will allow user to create an account by providing their name, password and email. The system will ensure access to user dashboard where they can design home interior, upload images, and use AI.
Source	Stakeholder Interview, Functional Requirement
Rational	Enables user to securely access personalized interior design features such as saving design, uploading room images, and AI based design.
Business rule	Email must be unique for each user. Password must meet security requirements.
Dependencies	Database, Authentication system
Priority	High

2.4 Non-Functional Requirements

These quality requirements should be specific, quantitative, and verifiable

Performance

The system should respond quickly and pages should load within 5-6 second.

Security

User data must be secured should be encrypted to prevent unauthorized access.

Usability

The system should be user friendly and easy to understand and use.

Reliability

The system should be stable with minimal errors or crashes.

Availability

The system should be available 24/7 for users.

2.4.1 Reliability

The home interior design and office focus on create a design without software failures frequency.

1. **Consistence performance**
   The system always gives correct design when the user designs the design and also AI based design.
2. **Data safety**
   User designs and uploaded photo are saved correctly without lose.
3. **Accurate output**
   The AI based design match user requirements like furniture style, room size, and colors.

2.4.2 Usability

Here are the Usability Requirements of the system:

1. **User friendly design tool**
   User can easily select furniture, colors, and layout without needing technical knowledge.
2. **Easy interface**
   The system has a simple and clear interface so users can easily navigate different design option.
3. **AI design feature**
   Users can generate interior designs quickly by giving simple inputs and uploading images.

2.4.3 Performance

Here are the Performance Requirements for system:

1. **Fast respond time**
   The system quickly loads pages and generates interior design result without delay.
2. **Efficient AI processing**
   AI generates a design for every time and also complex room designs are created.

2.4.4 Security

Here are the Security Requirements for the system:

1. **User authentication**
   Only registered users can log in access the design.
2. **Data protection**
   User personal information, uploaded images and saved design are securely stored.

3. **Secure login system**
   Passwords are protecting using secure methods to prevent unauthorized access.

2.5 External Interface Requirements

Here is the External Interface Requirements of the system:

1. **User interface**
   Simple and user friendly web interface for selecting furniture and colors and designs.
2. **Hardware interface**
   Works on laptops, desktops, and mobile devices with internet access.
3. **Software interface**
   Runs on web browser and uses database for example MySQL and AI services for designs.

2.5.1 User Interfaces Requirements

Here's a detailed description of the logical characteristics of each user interface that the system needs.

1. **General UI interface**
   - Simple and user friendly interface
   - Accessible design
   - Easy navigation between all features
2. **Users interface requirement**
   - Login/registration pages
   - Home dashboard
   - Design creation page
   - Image upload page for AI-based design
3. **Admin interface requirement**
   - Login page
   - Admin dashboard
   - Manage users and design
   - View and control AI-generated design request

2.5.2 Software interfaces

Here are the Software Interface Connections of the system:

1. **Database management system**
   - **Primary database:** MySQL for structured data and store user information, design and data securely.

2. **Frontend Frameworks & Libraries**
   - HTML, CSS, Java script for web interface development.
   - Bootstrap For responsive and beautiful UI design

3. **Backend Services**
    - **PHP**: Server side scripting to handle user login, data processing and backend.

2.5.3 Hardware interfaces

Here are the Software-Hardware Interface Requirements of the system:

1. **Supported Hardware Device Types**
    - Desktop, computer, laptop and smartphone to access the system.
2. **Output devices**
    - Monitor/screen for displayed interior design and result.
    - Printer (optional) for printing design layouts.

2.5.4 Communications interfaces

These are the Communication Requirements of the system:

1. **HTTP/HTTPs protocols**
    - Used for secure communication between frontend and backend server.
2. **Internet connection**
    - Required for accessing AI design features, database, and online services.

2.6 Summary

Summary of Requirements Identification and Use Case Analysis for home interior design and office system.

1. **Requirements Identification Process**

The key processes involved area:

- Collect both functional and non-functional requirements.
- Utilize standard formats like Use Case Diagrams

2. **Use Case Analysis Process**

The process includes:

- **Identifying Actors**: Define all user roles interacting with the system such as user and admin.
- **Developing Use Cases**: Create detailed use cases for each actor.



Chapter 3 Design and Architecture

3. System Design

These are system design constraints are as follow

Product perspective

The home interior design system is a digital tool. This tool helps the users to create their home Design select furniture and try color and build layout. This system has functionalities similar to existing interior design platforms in the market. But this system user-friendly and Interactive and using Ai.

System overview

Home interior design is a web based platform that facilitates this design home interior (rooms, kitchen, offices room etc.) user choice and also design with the help of AI. The system is minimize the time and create the design on this time.

3.1 Design consideration

Here are some Assumptions and Dependencies, Limitations, and Risks for the home interior design

System:

Assumption

- User provide valid information for login/signup
- User has basic Knowledge to use the website
- System assume that user upload rooms photo high quality then system extract
- accurate color palate
- System assume that user have stable internet connection
- Budget calculation feature does not include in this version but can be add in future

Dependencies

1. PHP and MySQL must be install and running for backend operations.
2. Three. js support is required for 3D preview functionality.
3. Color extraction library must function correctly for accurate palette detection.
4. User interface responsiveness depends on CSS and Bootstrap compatibility with user
5. Devices.

Limitations

- This plate form did not work without internet
- 3D may be slow on low-end mobiles or computer
- Color look different on every screen so may be exact color did not match perfect

Risks:

- Website may run slow or low-performance device during real time design
- Limited furniture option may reduce user creativity
- Interface does not working properly use face problem during drag-and-drop feature
- Website not working on same all the devices

3.2 Design Model

Provide the description of the following models used to describe the system design. Also ensure visibility of the all diagram.

1. Class diagram

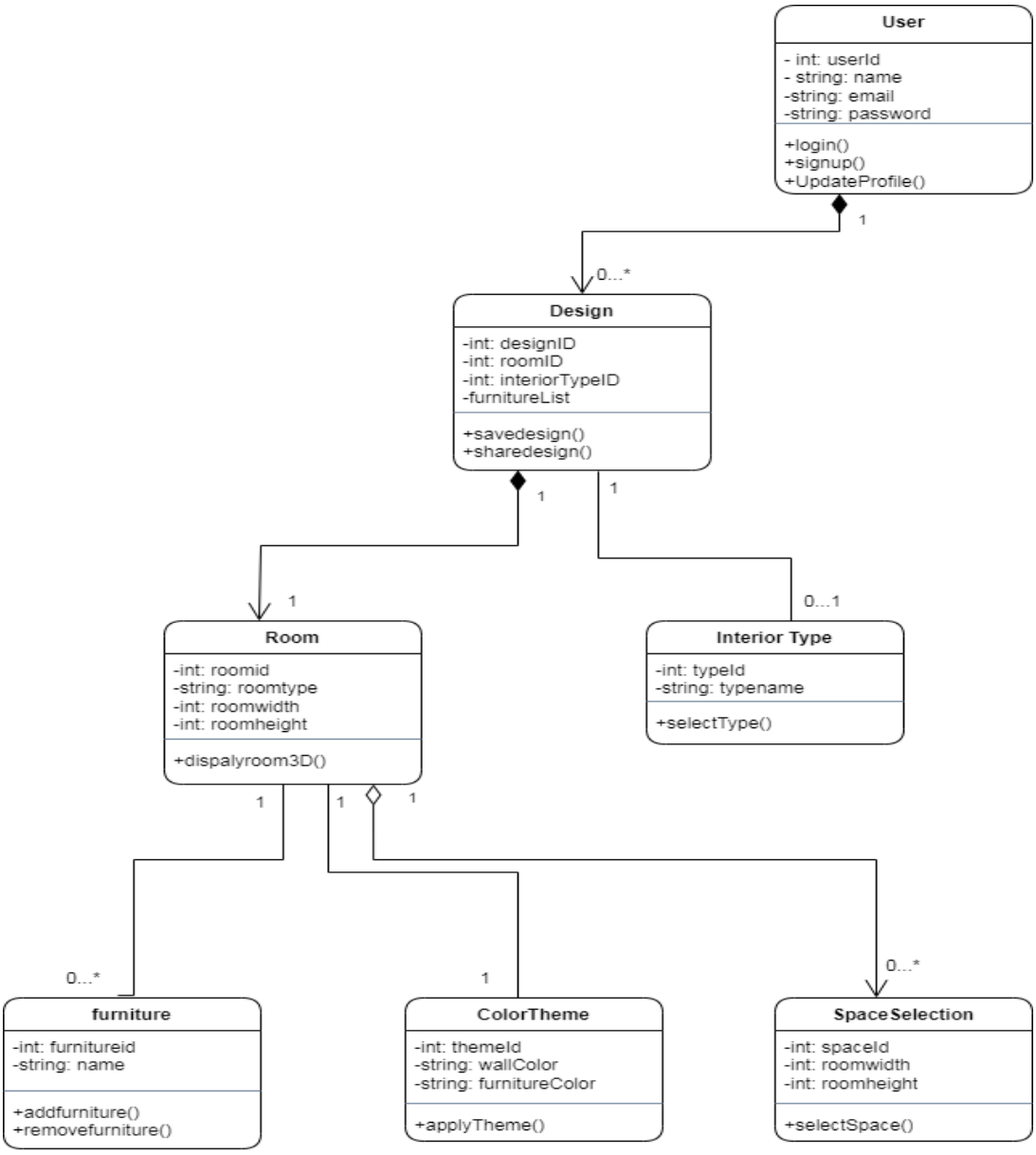


Figure 3 class diagram

Sequence diagram

Use Case UC_1: UC_login\register

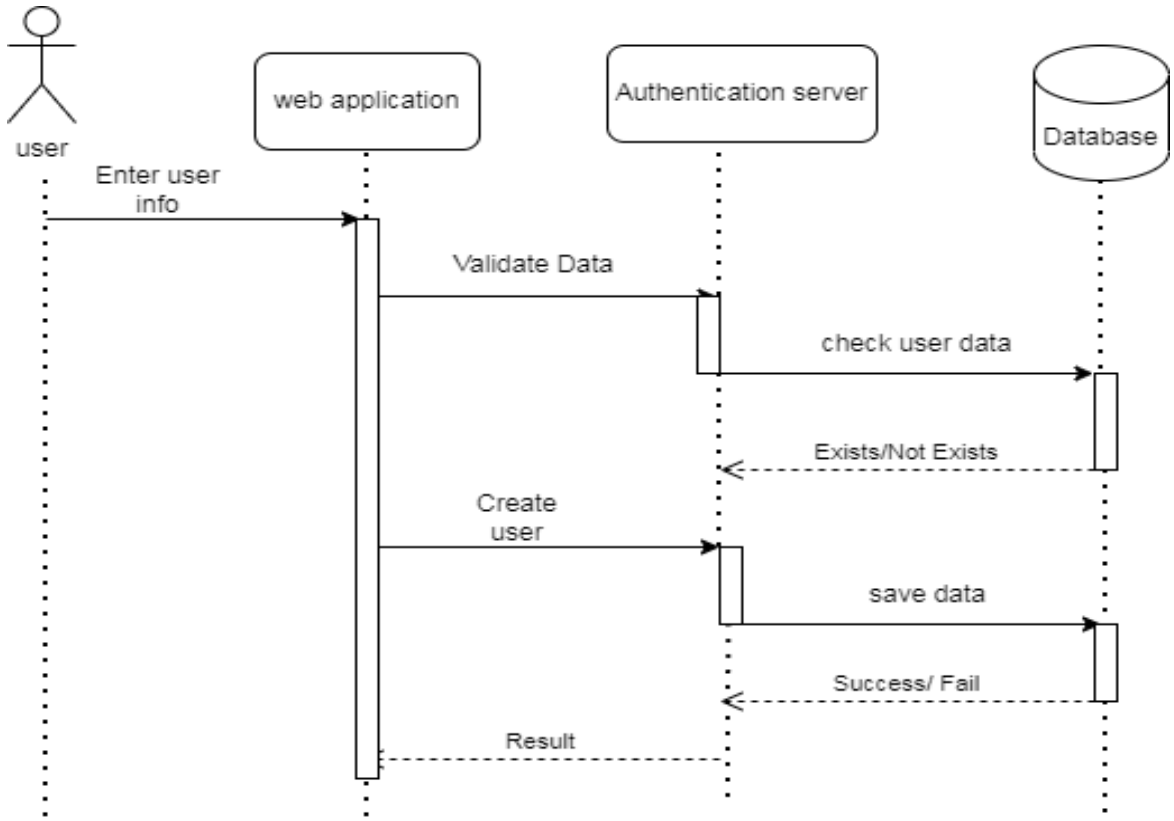


Figure 4 Sequence Diagram UC 1

Use case UC_2: UC_Selection module

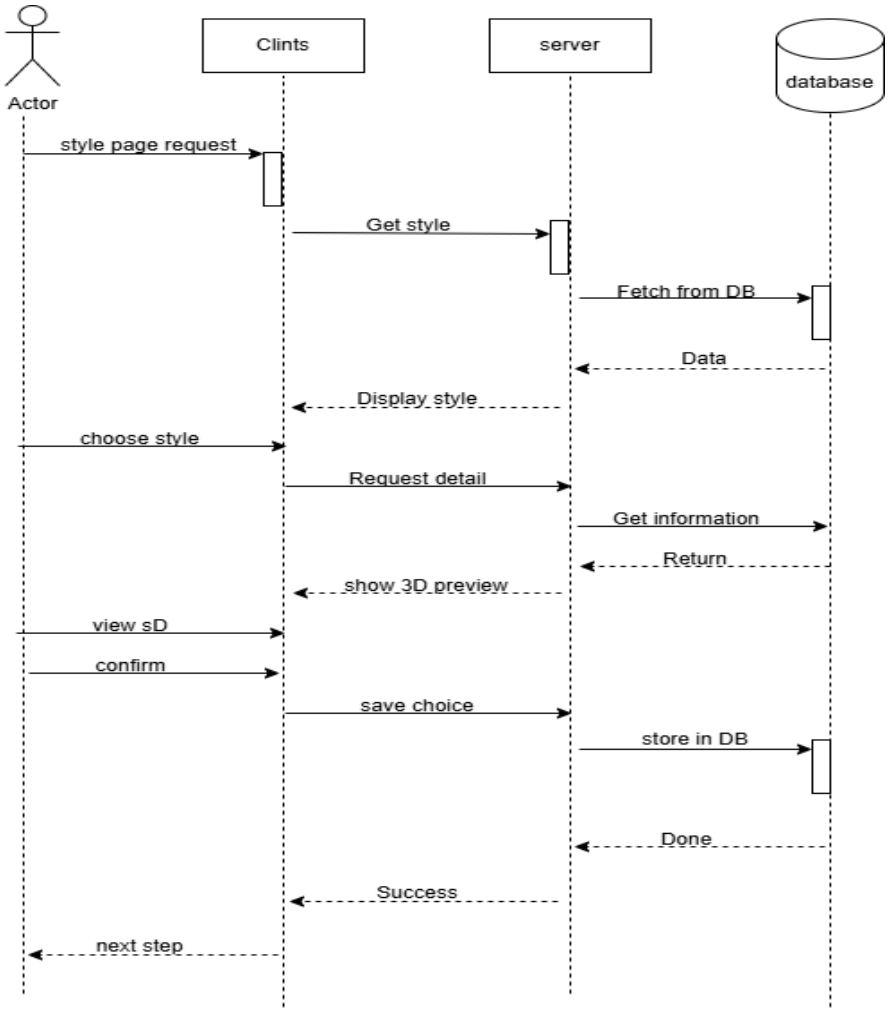


Figure 5 Sequence Diagram UC 2

Use case UC_3: UC_Room type selection

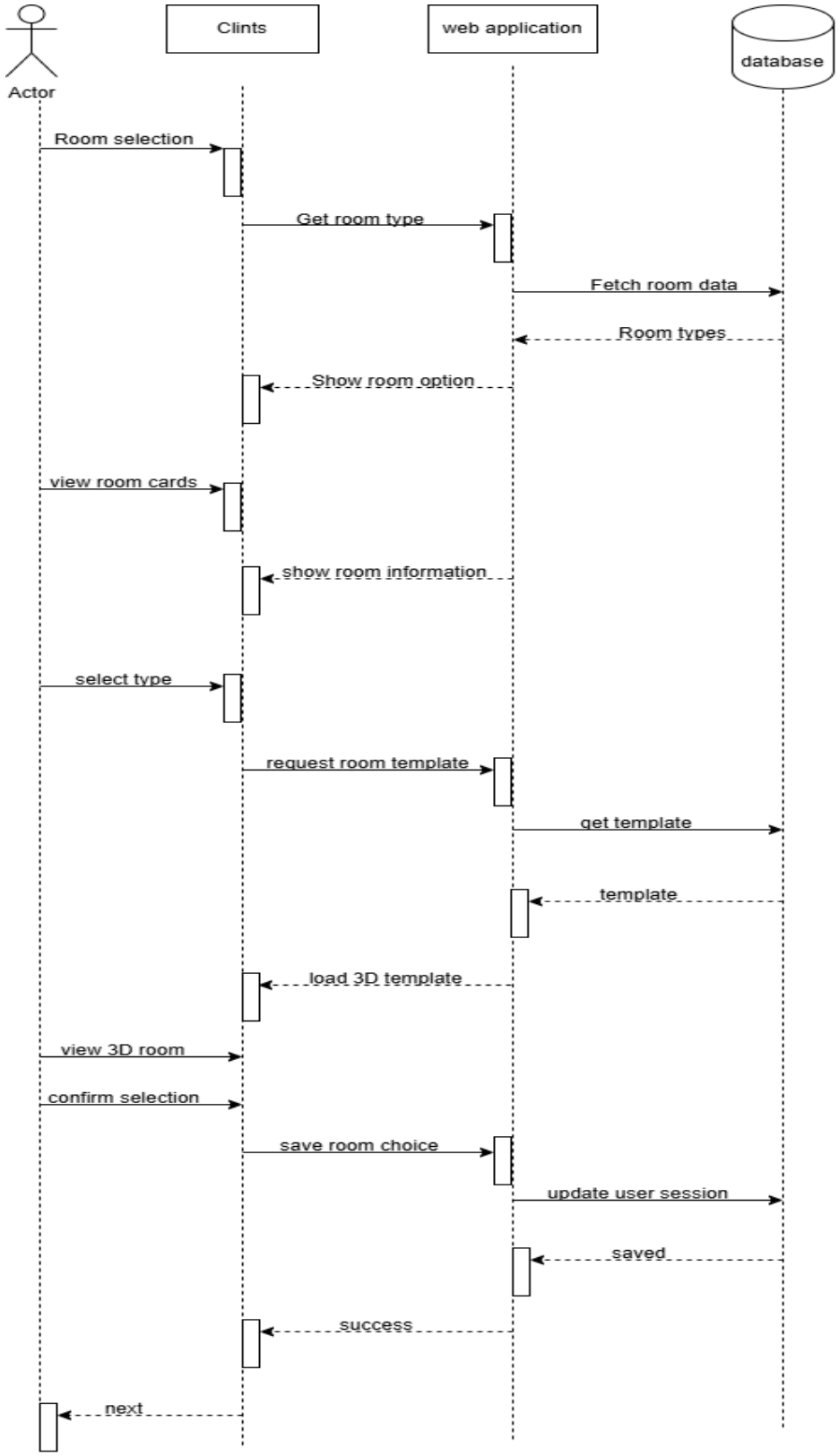


Figure 6 Sequence Diagram UC 3

Use Case UC_4: UC_ upload photo

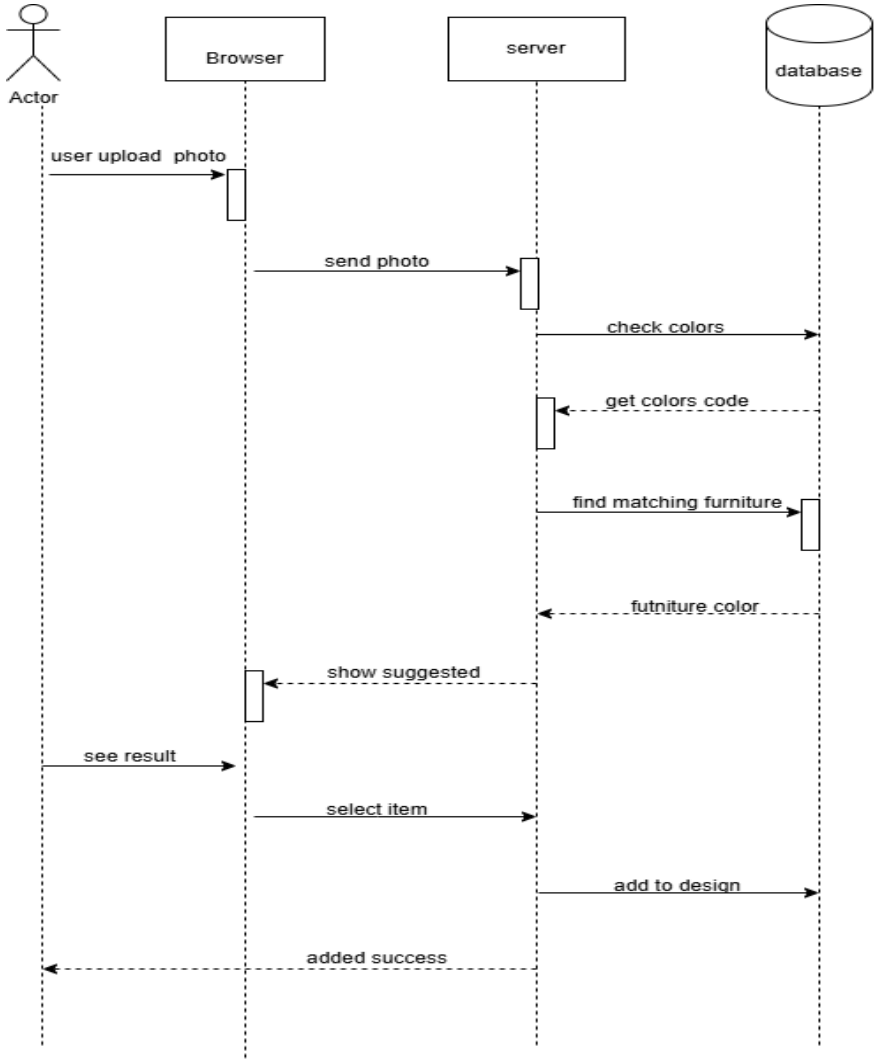


Figure 7 Sequence Diagram UC 4

Use Case UC_5: UC_Save and share

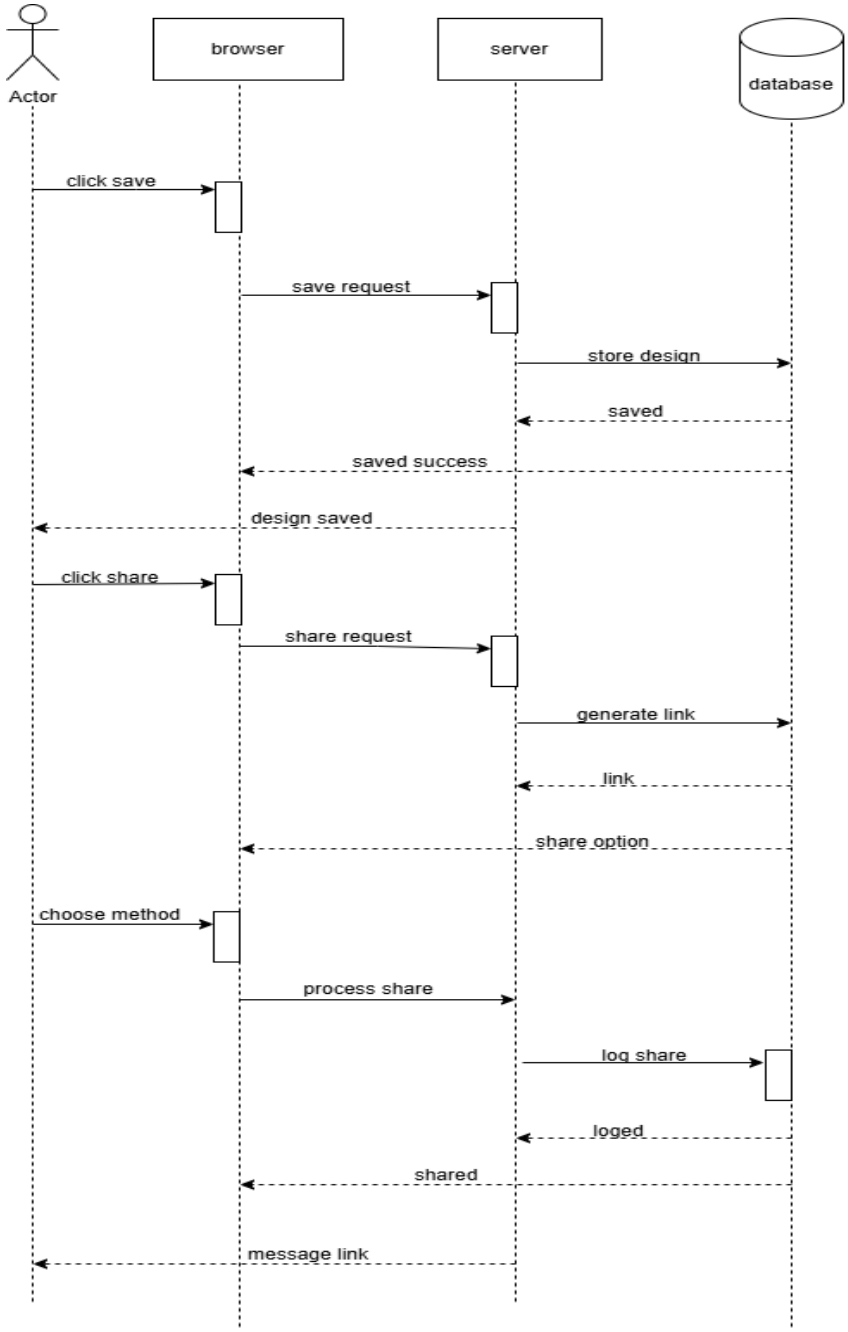


Figure 8 Sequence Diagram UC 5

2. ER Diagram

It captures the key entities, attributes and relationship between them.

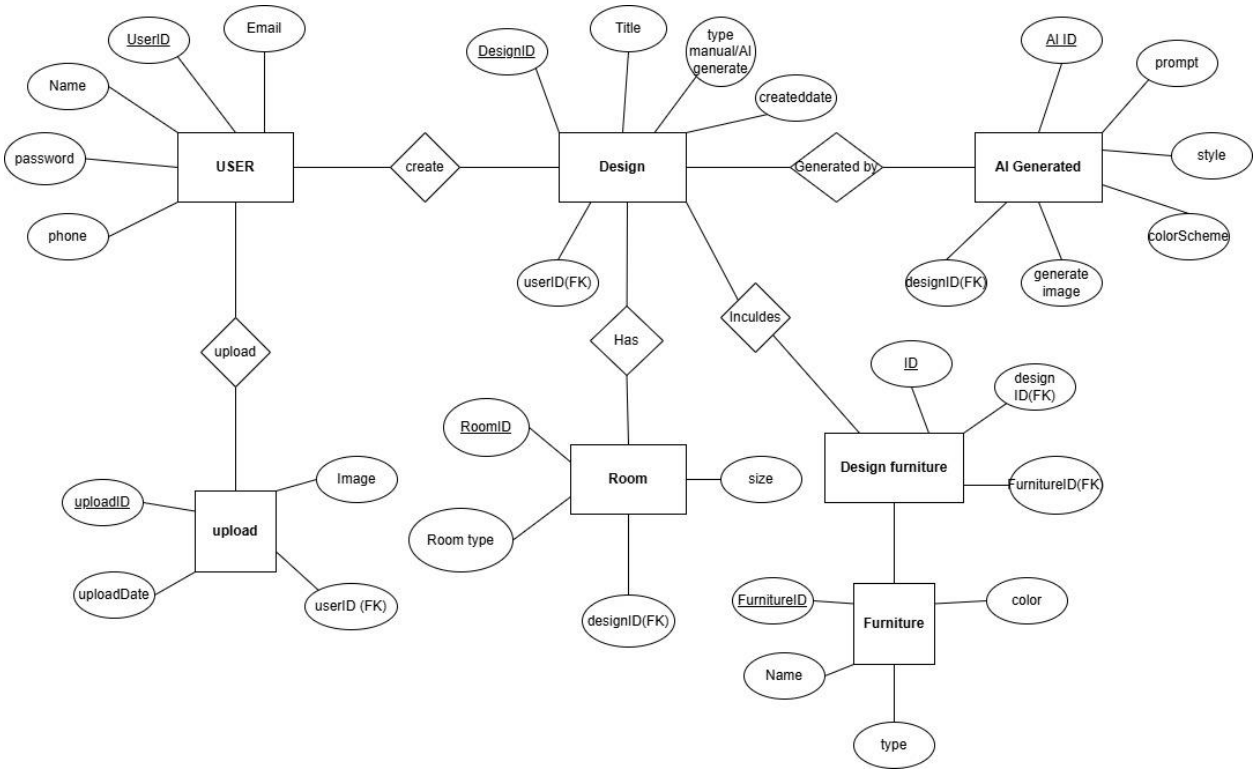


Figure 9 ER Diagram

3.3 Architectural design

- **UML class relationship diagram**

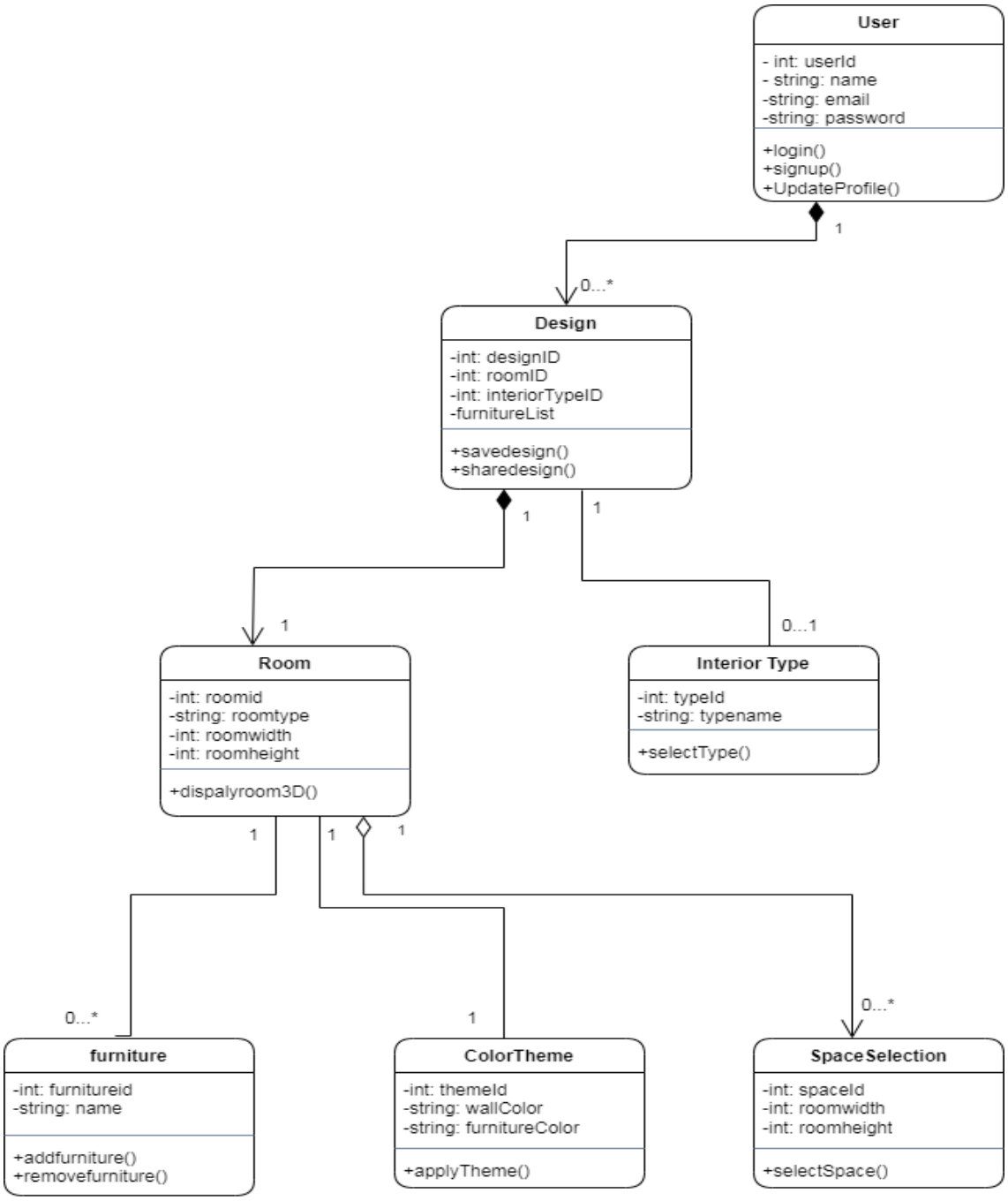


Figure 10 Class Relationship Diagrams

- **UML component diagram**
  Design Module is what user sees and interacts with furniture catalog in a browser. The material module work behind the scenes to manage all data like item detail by talking database to save and fetch information.

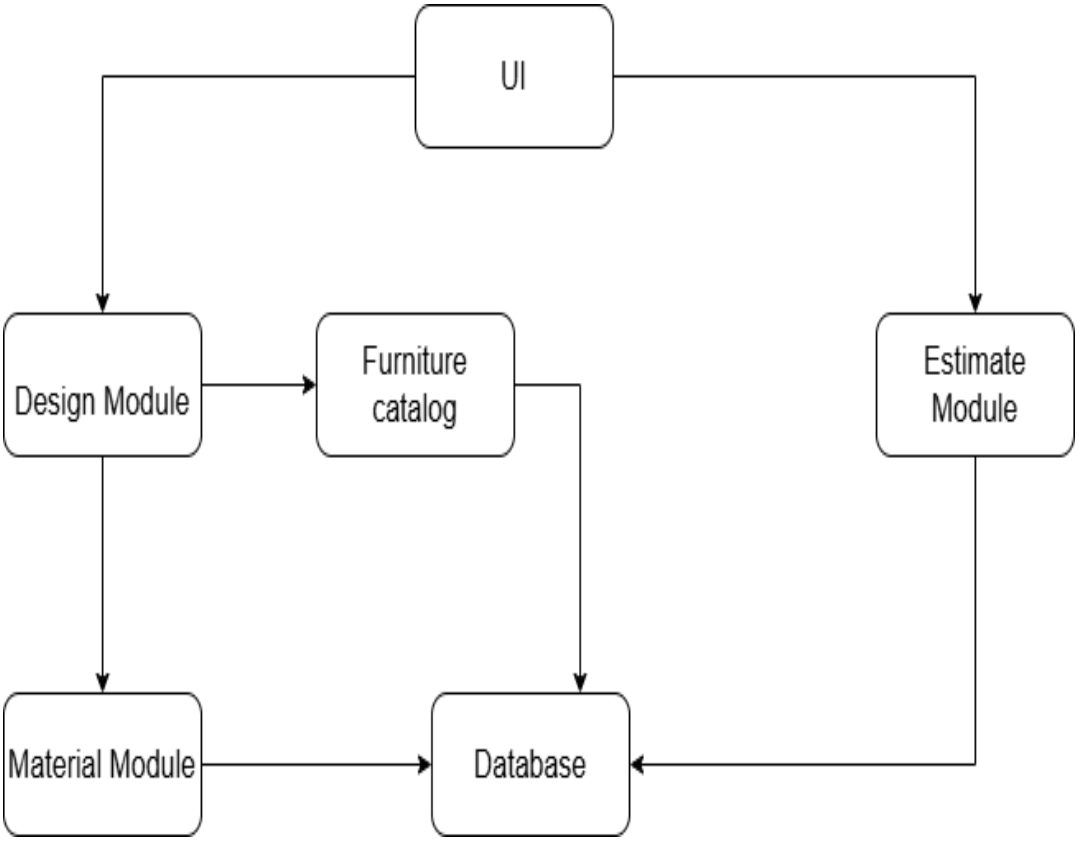


Figure 11 UML Component Diagram

MVC (Model view Controller)

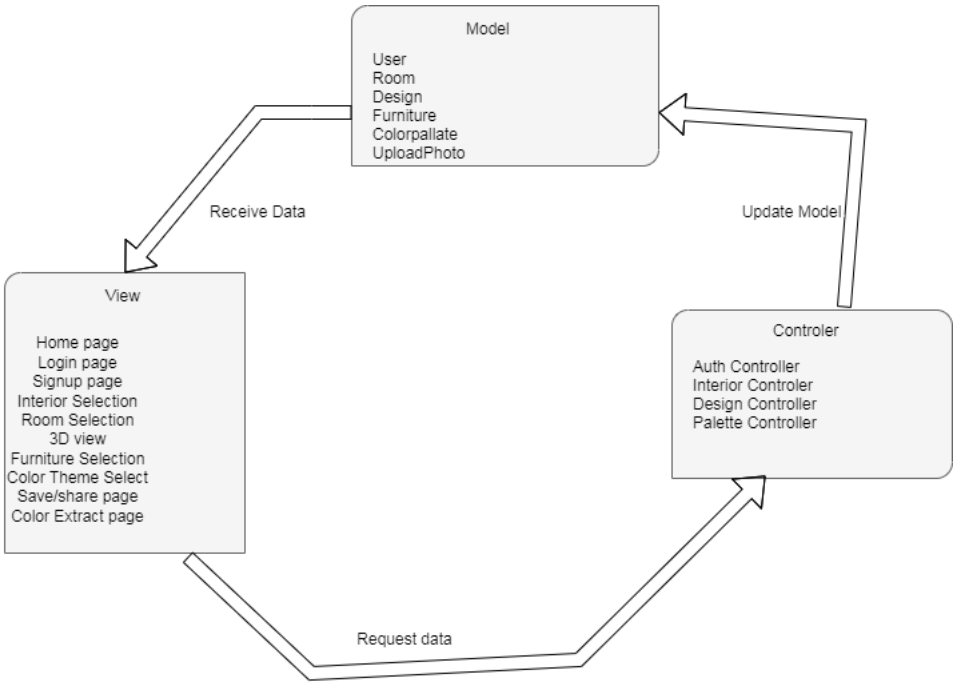


Figure 12 MVC diagram

3.4 Data design

Here an explanation how the information domain of the interior design home and office system is transformed into data structures:

Information Domain Transformation

The information Domain of the system consists of user, rooms, design, furniture and interior type. These entities are transformed into data Structure is as follows.

- **User**
  - User ID (Unique Identifier)
  - Name
  - Email
  - Password
- **Room**
  - Room ID (Unique Identifier)
  - User ID
  - Room Type (Bedroom, office, living room)
  - Width
  - Height
- **Furniture**
  - Furniture ID (Unique Identifier)
  - Name
  - Category
  - Dimension

- **Interior Type**
  - Interior Type ID (Unique Identifier)
  - Type name (classic, modern)
- **Design**
  - Design ID (Unique Identifier)
  - Room ID
  - Interior Type ID
- **Room photo**
  - Photo ID
  - Room ID
  - Image path
  - Upload Date
- **Design_Furnitue (Mapping Table)**
  - Design ID
  - Furniture ID

Data Structure

The system uses a relational database management system to store and manage data.

The following data structures are used:

Tables

- **User**
  - User ID (primary key)
  - Name
  - Email
  - Password
- **Design**
  - Design ID(primary key)
  - Room ID(Foreign key)
  - User ID(Foreign key)
  - Interior Type ID(Foreign key)
- **Room**
  - Room ID (primary key)
  - User ID (Foreign key)
  - Room Type (Bedroom, office, living room)
  - Width
  - Height
- **Furniture**
  - Furniture ID (primary key)
  - Name
  - Category

- Dimension
- **Interior Type**
  - Interior Type ID (Foreign key)
  - Type name (classic, modern)
- **Design_Furnitue (Mapping Table)**
  - Design ID (Foreign key)
  - Furniture ID (Foreign key)
- **Room_photo**
  - Photo ID
  - Room ID
  - Image path
  - Upload Date

Relationship

1. A user can have multiple rooms (one-to-many).
2. A user can have multiple designs (one-to-many).
3. A design belongs to one room and one interior type (many-to-one).
4. A design can have multiple furniture items through design furniture (many-to-many).
5. Furniture item can be part of multiple designs (many-to-many).

3.4.1 Data dictionary

Alphabetic list of system entities or major data with type and description.
The following table presents an alphabetic list of major entities, attribute and data items. Each entry include in data type and a brief description of its purpose within system.

Attribute	**Description(Data Type & purpose)**
Category (Furniture)	VARCHAR – Describe the category of furniture(chair, office table,sofa)
Design	Entity – Represent a complete interior design associated with a room and furniture.
Design ID(PK)	INT – unique id assign to every design
Design Furniture	Mapping Entity - Connect a design to selected furniture item
Dimension (Furniture)	VARCHAR – Length, width, height specification of furniture.
Email (User)	VARCHAR – unique email used for login
Furniture	Entity – Store all available furniture items for selection.
Furniture ID (PK)	INT – unique identifier for furniture item.
Furniture ID (FK)	INT – Foreign key referencing the Furniture table.
Height (Room)	INT – Height measurement of the room in chosen unit

	feet/meters.
Interior Type	Entity – Define the style of a design (modern, traditional, office style)
Interior Type ID (PK)	INT – primary key for interior type entity
Interior Type ID (FK)	INT – Foreign key linking Interior Type to design
Name (Furniture)	VARCHAR - Name / title of furniture item (e.g "wooden chair)
Name (User)	VARCHAR – User full name
Password	VARCHAR – Encrypted password used for user login authentication.
Quantity (Design Furniture)	INT – Number of units of a particular furniture used in a design.
Room	Entity – Represent the physical user room designed.
Room ID (PK)	INT – primary key unique identifier of each room
Room ID (FK)	INT – Foreign key referencing the room table
Room Type	VARCHAR – Type category of room (bedroom, kitchen, office, living room)
User	Entity – Represent an individual who access the system
User ID (PK)	INT – unique identifier assign to each user
User ID (FK)	INT – Foreign key link to user room
Width (Room)	INT – width measurement of room

- **Structure Approach Function and Function parameter**

  For system following structured programming paradigm list all function along their function parameter. For each function providing a description detailing its functionality and for each parameter specify a data type and its role within function.

  Here list of function interior design home and office system along with their parameter, description and data types:

  1. **Login**
     - userEmail (string): email entered by user.
     - userPassword (string): password entered for verification
     - Description : Authenticate the user by checking email and password for stored records

  2. **Signup**
     - Name (string): new user full name
     - Email (string): new user email
     - Password (string): new account password
     - Description: Register a new user by storing profile detail in database

3. **Update Profile**
    - userID (int): identify which user update
    - newName (string): update name
    - newEmail (string): update email address
    - Description: update user profile information

4. **Save Design**
    - designID (int): unique design identifier
    - userID (int): user who created the design
    - roomID (int): selected room
    - InteriorTypeID (int): select interior theme
    - FurnitureList (list): list of furniture item
    - Description: save the design configuration created by the user into system

5. **Share Design**
    - Design ID (int):
    - Platform Name (string): platform (Facebook, email)
    - Description: Allow sharing the completed design to social platform

6. **Display Room 3D**
    - Room ID (int): identify the room
    - Width (int): room width
    - Height (int): room height
    - Description: Display the 3D room view

7. **Select Type**
    - Type ID (int): interior Style ID(modern, classic etc).
    - Description: Apply design style to current design

8. **Add furniture**
    - Furniture ID (int): furniture identifier
    - Name (string): name of furniture (sofa, chair etc)
    - Description: Add furniture item to selected room design

9. **Remove furniture**
    - Furniture ID (int): ID of furniture remove
    - Description: Remove the selected furniture

10. **Apply Theme**
    - Theme ID (int): selected theme

- Wall color (string): color code for wall
- Furniture color (string): color code for furniture
- Descrption : Apply a color theme on room and furniture

11. Recommend color For Image

- Image path (string): image location
- Description: Recommend color for upload image

12. Select space

- Space ID (int): space identifier
- Room width (int): width of space
- Room height (int): height of space

3.5 User interface Design

Here an overview of functionality of the interior design home and office from user perspective:

1. Overview From User perspective

The system provide an easy to use interface that allow user to design interior space home or office, select furniture, apply interior style and view the final room layout. The UI focus on simplicity and smooth navigation so that user with no technical background can easily complete all major tasks.

2. How user interact with the system

a) Home page

- Display system title and brief introduction
- Provide main navigation button
  - Login/signup
  - Create new design
  - Saved Design
- From here user can start a design or access saved work

b) User Registration and login

- User can create account by providing basic detail
- After login user access dashboard
- Feedback:
  - Success: Account created successfully
  - Error: Incorrect password or user already exist

c) Dashboard

- Create new design
- Continue existing design

- Manage profile
- Logout

d) Create new Design page

User can:

- Select room type (bedroom, office, kitchen etc)
- Choose interior style (modern, classic etc)
- Enter basic room dimension (length, width, height)
- Start designing the layout

e) Room editor interface

User can:

- User can drag and drop furniture onto room layout
- Resize, Rotate
- Change floor type, wall color or interior theme
- Preview the design in 3D
- Feedback:
- Furniture added to room
- Object rotated to room
- Furniture cannot overlap
- Design saved successfully

f) Furniture selection panel

- A side bar or pop up display:
- Furniture categories (sofa, office table, chair, cabinets, beds)
- Item detail (size, color, type)
- Add button item placing in room
- Feedback: select furniture added to workspace

g) Design preview

User can:

- View the complete room layout
- Switch between 3D render

h) Save, update and delete design

Button available:

- Save design
- Update existing design
- Delete Design

System conformation:

- Design saved successfully
- Design updated

- Design deleted

i) View saved Design

User can view:

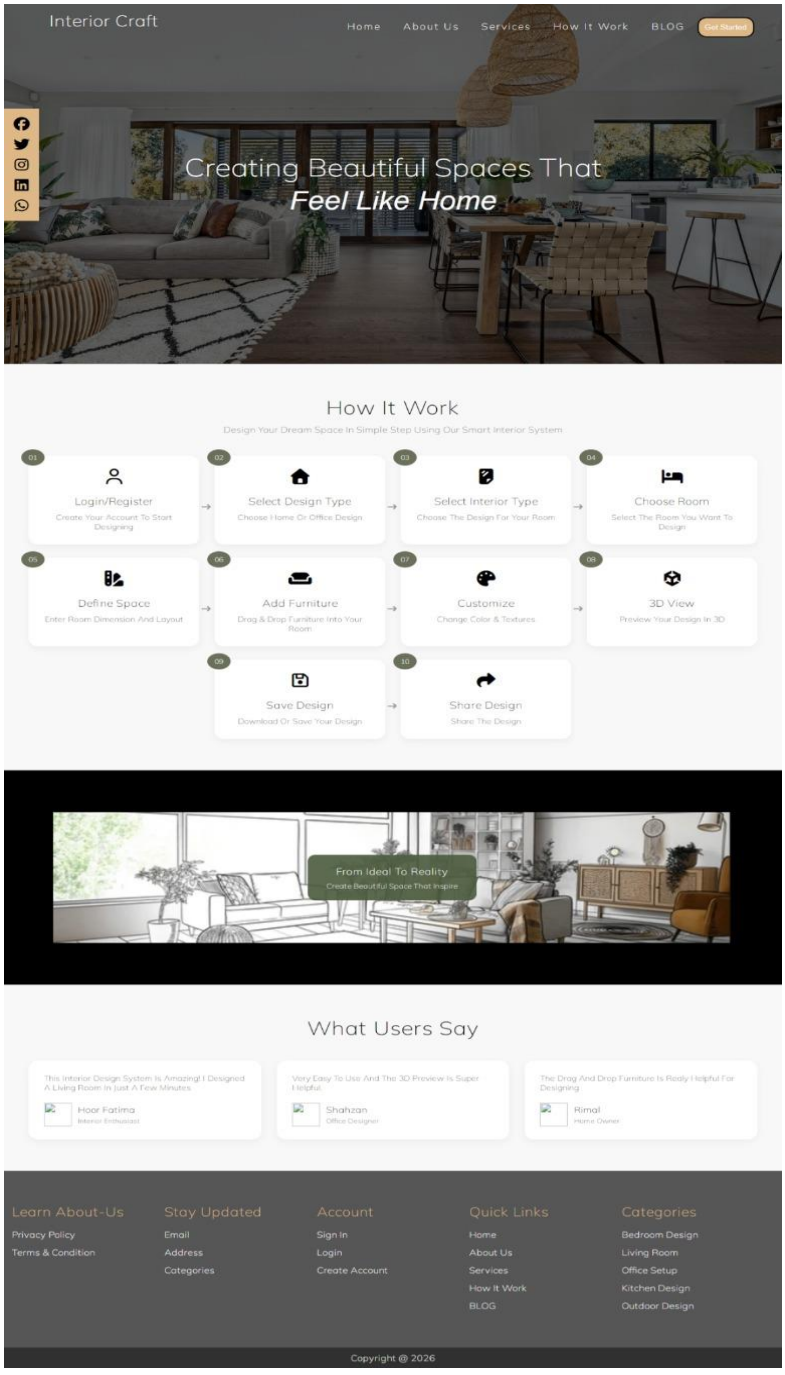
- All previously saved design
- Edit, export or continue working on any room

j) Logout

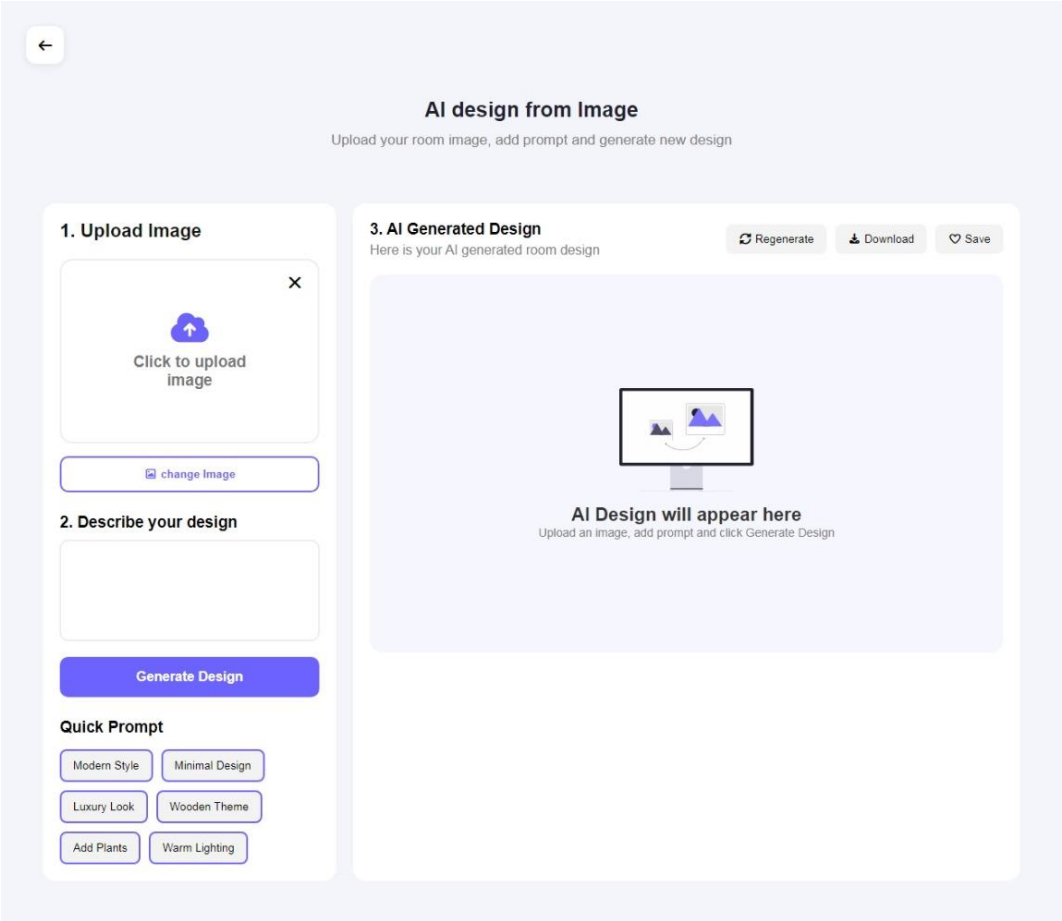
Return home page

3.5.1 Screen images

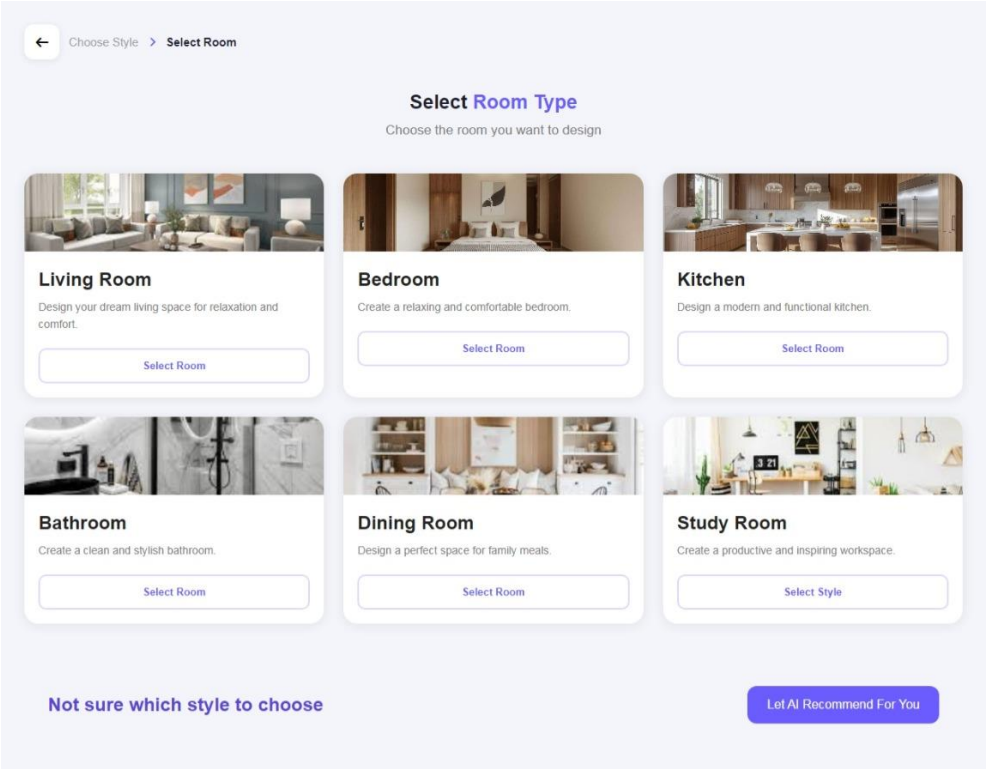
Home page



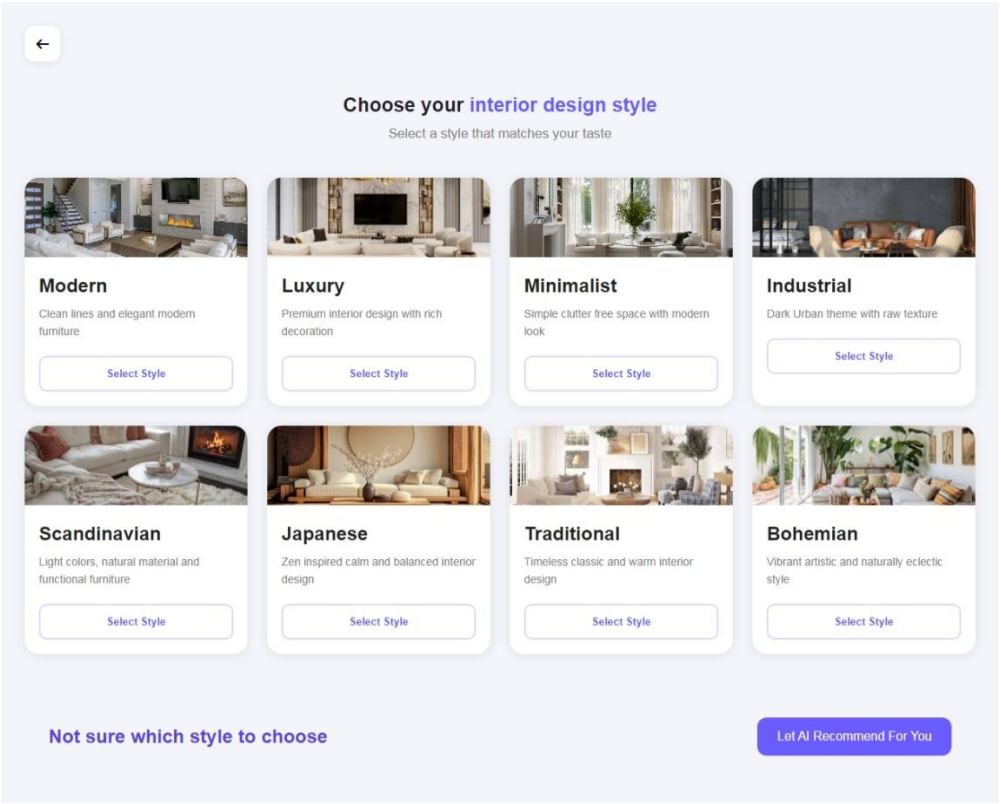
Upload a photo



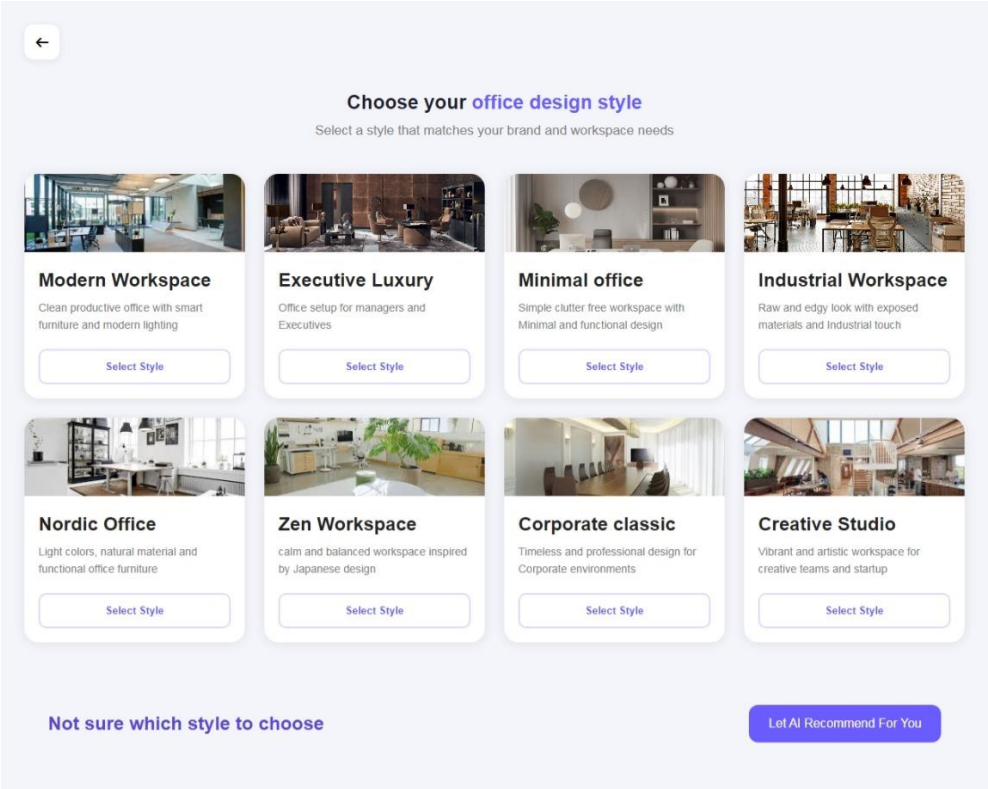
Selection Area



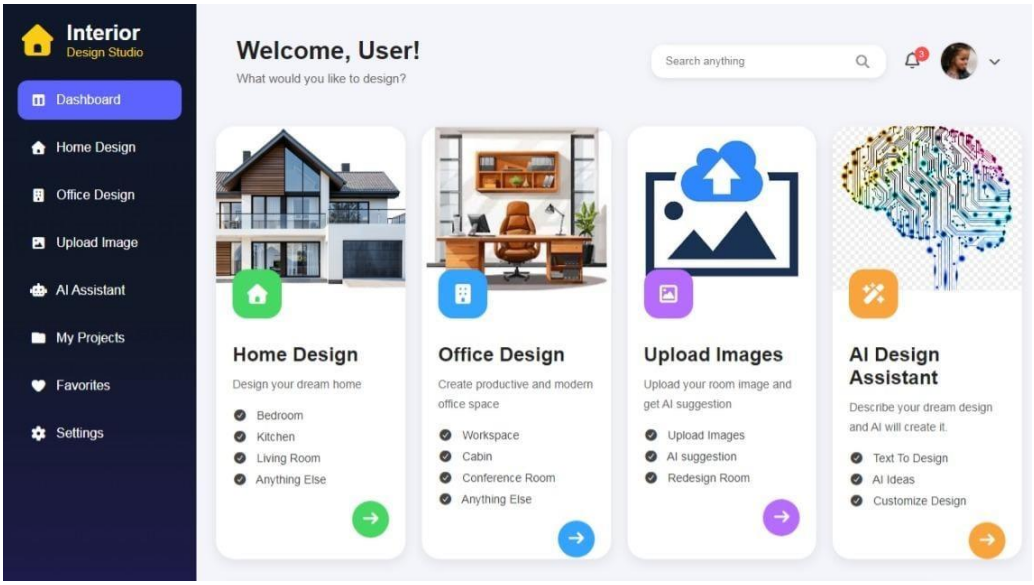
Select Design



Select office design



Create Design



3.6 Behavioral Model

State Transition Diagram

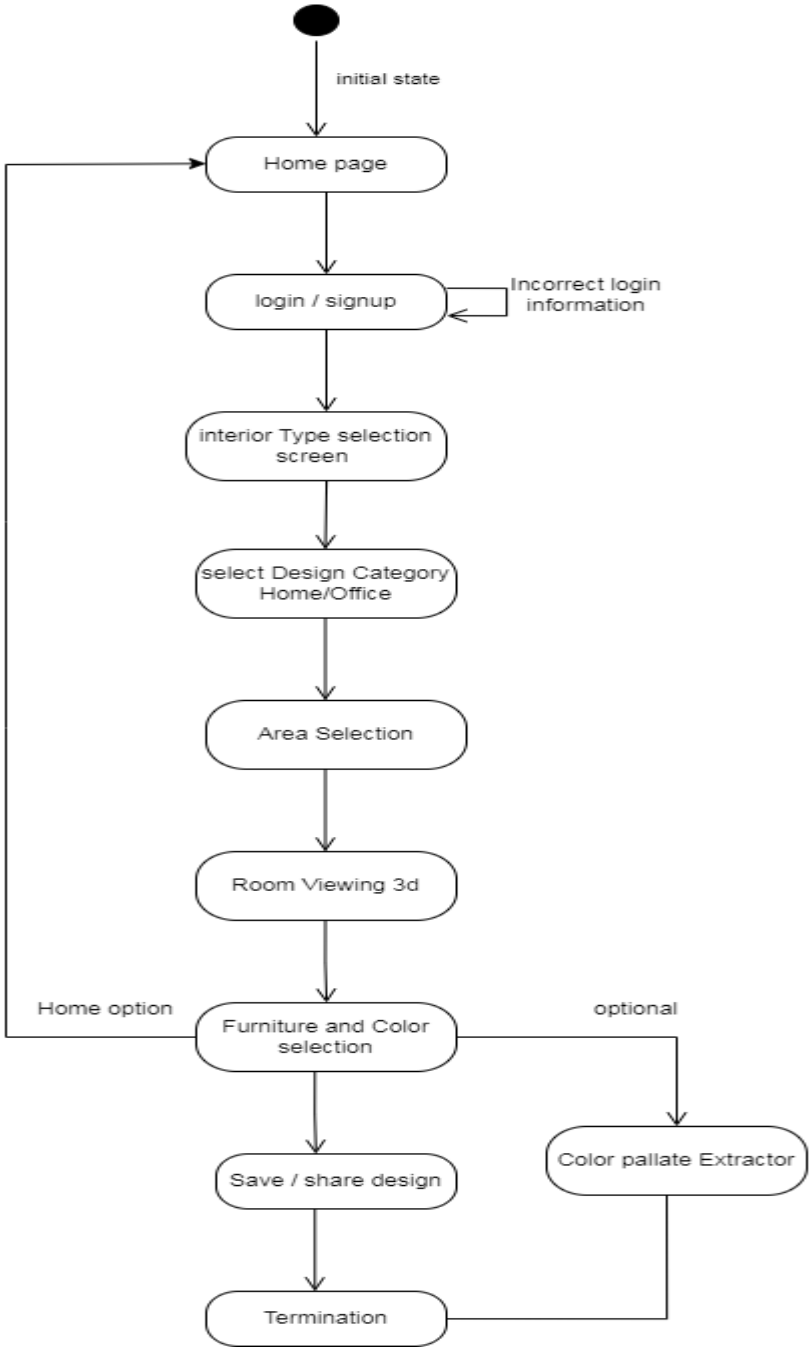


Figure 13 State Transition Diagram

3.7 Design Decision

Highlight the design choice while designing your system (e.g. choosing the object oriented design pattern, choosing between the algorithm approaches, normalization of the database etc.). Clearly specify which approach you have used and why you have made a specific design choice.

Here we can explain the design choice while developing Home interior design and office system.

Choice of Architecture – MVC Model

The system uses the Model-View-Controller (MVC) architecture.

Reason:

- Separate data, interface and logic for easy maintenance
- Support scalability as project grows
- Improve reusability of component (Model for data, View for UI, Controller for request handling)

Design Approach – Object Oriented Design

The system follows object oriented design (OOD) principles

Reason:

- Fits naturally with real world entities (user, room, design, furniture)
- The system built in in small parts (e.g. adding new room type or furniture item)
- One piece of code can be used again in different places

Algorithm choices

3D Rendering: Three.js selected for 3D room preview because it is lightweight and work well in browser without installing software.

Color palate extraction: A simple color detection algorithm/library is used to achieve fast processing without heavy computation.

Reason:

- Improve performance and reduce on low end devices
- Ensure browser compatibility

Database design decision

The database uses Relational Database Model (MySQL)

Reason:

- Structured data with strong relationship (user-design-room-furniture) with relational tables
- Ensure data integrity using foreign key
- Easy to implement and maintain for web application

Normalization level

Database table follows 1NF-2NF-3NF

Reason:

- Remove data Redundancy
- Ensure accurate and consistent data storage
- Improve query performance and reduce update problem

Technology stack decision

Frontend: HTML, CSS, Bootstrap, JavaScript

- Chosen for responsive design and simplicity

Backend: PHP

- Easy to integrate with MySQL and widely supported

3D Engine: Three.js

- Required for browser based 3D visualization

Database: MySQL

- Reliable, easy to manage, and support relational modeling

Interaction design choice

A drag-and-drop interface for room designing

Reason:

- Make designing easy for non-technical user
- Make furniture and layout customization simple

Performance and usability decision

- We made the 3D model lighter so the website load faster and does not slow down
- We used common color formats so the color look almost the same on ever mobile, laptop are screen
- We used small and simple software tools so the websites work smoothly even on old are low power devices

Summary

This chapter explained the main design idea used in the interior design home and office system. It highlights how the system is structured, how the data is organized, and what design decision was taken to make the system easy to use, scalable, and efficient. It describe how the system architecture, database design, object oriented approach and user interface choice were planned to meets the goal of the project. The use of the MVC pattern, relational database, modular design help the system stay organized, easy to update and simple to maintain.

The chapter also showed how different components such as 3D preview, color extraction, furniture catalog, and user interaction flows work together provide a user friendly experience. These design decision ensure that the system is efficient, flexible and able to support future improvement.

Overall the design choices are important in this chapter because they create a strong foundation for developing a functional, reliable and easy to use interior design application that meet the need of user and goal of the project



Chapter 4 Implementation

4. Implementation

The chapter discusses the implementation details of the Home Interior Design and Office. The core component are described the below in pseudo code from, covering the major functional modules of the system.

4.1 Algorithm

Table 7 Examples of Algorithm

Algorithm 1- User Authentication

Algorithm	User Authentication
Input	Email , Password
Output	Access Granted/ Denied
Pseudo Code 1: START 2: INPUT email, password 3: Retrive user record from database 4: IF email exists THEN 5: IF password matches THEN 6: Display Login Successful 7: ELSE 8: Display Invalid Password 9: ELSE 10: Display User Not Found 11: END	

7.1 Table 1 Algorithem-1

Algorithm 2- Search and Filter Design

Algorithm	Search and Filter Design
Input	Search keyword(e.g. “sofa”,”modern”)
Output	List of matching Design
Pseudo Code 1: START 2: INPUT Search keyword 3: FOR each item in design_list DO 4: IF item matches Search keyword THEN 5: Add item to result_list 6: Display result_list 7: END	

7.2 Table 2 Algorithem-2

Algorithm 3- Room Layout placement

Algorithm	Room Layout placement
Input	Room Dimensions, Selected furniture
Output	Arranged Layout
Pseudo Code 1: START 2: INPUT room_width, room_height 3: INPUT furniture_list 4: Initialize empty layout grid 5: FOR each furniture in furniture_list DO 6: Find available position in grid 7: Place furniture at position 8: Display final layout 9: END	

7.3 Table 3 Algorithm-3

Algorithm 4- Recommendation System

Algorithm	Recommendation System
Input	User selected room type/style
Output	Suggested design
Pseudo Code 1: START 2: INPUT user_preferences 3: FOR each design in database DO 4: IF design category == user_preferences THEN 5: Add design to recommendation_list 6: Display recommendation_list 7: END	

7.4 Table 4 Algorithem-4

Algorithm 5- 3D View Rendering

Algorithm	3D View Rendering
Input	Selected furniture object/ room dimension
Output	3D visualize room design
Pseudo Code 1: START 2: Initialize 3D scene 3: Set camera position 4: Add lighting to scene 5: FOR each object in furniture_list DO 6: Load 3D model of object	

```
7:        Set position (x,y,z)
8:        Add object to scene
9:   Render scene on screen
10:  END
```

7.5 Table 5 Algorithem-5

4.2 External APIs/SDK

The system integrates multiple third-party APIs and SDK to enhance functionality. The table below provides a detailed breakdown of each external dependency used in project implementation.

Table 8 Details of API used in project

API/SDK Name & Version	**Description**	**Purpose of usage**	**List down the Api endpoint/function/class in which it is used**
Three.js	A JavaScript library used to create and display animated 3D graphics in the browser using WebGL	Used for rendering 3D room design, place furniture, camera view, lighting	Scene(), Camera(), Mesh(), Light(), Renderer()
WebGL	Low-level graphics API for rendering 3D graphics in web browser	Provide hardware accelerated rendering support for 3D visuals	RenderingContext, shader, buffer
Bootstrap	Frontend Framework for responsive design and UI component	Used for designing form, navigation bar, layout, button	Grid system , Navbar, form, button
HTML5 File Api	Allow user to upload and read file from their system	Used for uploading room images	File Reader(), Input type=file

4.3 Code Repository

To manage version control and collaboration effectively for group project, Git is used as primary tool. All project file include code, documentation and other relevant resources are stored in Git Repository. This ensures proper tracking of changes,

collaboration among team member, and access to the most up-to-date version of the project.

Git Repository Link

The repository for group project can be accessed using the following link

https://github.com/javairia-maker/Home-interior-design-and-office

4.3.1 Metrics of the Git repository

The following metrics are monitored to ensure effective use of git repository

- **Commits:** represent the number of changes saved in repository. In this project commit are used to track update such as UI design. Regular commits ensure that progress is consistently recorded.
- **Branches:** are used to manage different feature of the system. E.g. Login system, 3D room design. These branches are later merged into main branch after completion.
- **Pull Request:** are used when a developer wants to merge changes from one branch to another. In this project pull request help to reviewing new feature like layout design and search functionality before adding them to main system.
- **Issues:** are used to track bugs, errors and feature requests. E.g. Improving 3D rendering performance, adding new feature categories. Issues help in organizing and managing project task effectively.
- **Contributor:** are the team members working on the project. Each contributor work is tracked based on number of commits, code addition and code modification.

- **Code Review:** are conducted before merging any changes into the main branch. This ensure code quality in maintained, Errors are minimized.

4.4 Summary

This chapter presented the implementation of the Home Interior design & Office system where core feature such as user authentication, search and filter, room layout, recommendation system and 3D view render are developed using algorithm and pseudo code. Technologies like Three.js, WebGL are used for 3D visualization, while bootstrap ensure a responsive UI and HTML5 FILE API support image uploads. Git is used for version control to manage code, track changes and team collaboration.



Chapter 5 Implementation

5. Introduction

To ensure effectiveness of software verification and validation all test script and cases should confirm to the following best practice.

- **Keep tests independent:** Each furniture should be tested separately so one failure does not affect others.
- **User meaningful assertions:** Validate outputs clearly such as checking if the AI design is generated correctly or if upload images match furniture suggested.
- **Handle test data:** use proper setup and teardown methods to manage user data design files and uploaded images to keep the system clean.
- **Test repeatable:** ensure that test for design generation login system and image uploaded give consistent result every time they are executed.

5.1 Unit Testing (UT)

Unit testing involves testing individual components or function in isolation. The goal is to ensure that part of the system works correctly on its own.

Example: verifying that the designGenerater () functions correctly creates a room design based user input and returns the expected layout and recommendation

Table 9 unit testing

Test case ID	**UT**
Requirement ID	REQ-login-001
Title	User login function
Description	Validate user login with correct credentials
Objectives	To ensure successful authentication of user in system
Driver /precondition	Database must be running; user account must be already exits.
Test step	1.Open postman 2.Send POST request to http://localhost:4000/api/admin/login enter email and password

	click enter button
Input	{"email":user@gamil .com,"password"123456.
Expected result	User login successful authentication token generated
Actual result	Token returned successfully
Remarks	Pass

5.2 Functional Testing (FT)

Functional testing ensures that each feature of the system works according to user requirement. It verifies that all functionalities like login, AI generated design, and image upload and furniture matching are working correctly from the user perspectives.

Example: ensuring that only logged –in user can access the design dashboard and generate AI-based interior design while unauthorized users are restricted from accessed these features.

Table 10 functional testing

Test case ID	**FT**
Requirement ID	REQ-2
Title	AI interior design generating
Description	Users can generate interior design based on their preference
Objectives	Ensure user account can successfully create design using AI
Driver /precondition	Valid user account and user must be logged in
Test step	1. Login as user 2. Enter room detail (type, color, style) 3. Click “generate design”
Input	User credentials +design input(room type , color, style)
Expected result	Design is generated and displayed with matching furniture
Actual result	As expected
Remarks	Pass

5.3 Integration Testing (IT)

Integration testing focuses on checking how different modules of the system work together. It ensures that all components such as user login, AI design generation, image upload, and furniture matching interact properly and produce correct results as a complete system.

Example: testing if a user login session correctly connects with the AI design module, allowing the user to generate and save interior designs, and ensuring that uploaded images are properly used by the furniture matching system to suggest suitable design.

Table 11 Integration testing

Test case ID	**IT**
Requirement ID	REQ-3
Title	Save design→retrive design
Description	Test newly created AI design is properly saved and later displayed in user design list
Objectives	To confirm data flow from design generation →database storage→retrival in dashboard
Driver /precondition	User must be login
Test step	1. Login as a user 2. Generate AI interior design 3. Click save design 4. Go to “my design” section
Input	User details+AI design input
Expected result	Save design appears in user design list with correct details
Actual result	As expected
Remarks	Pass

5.4 Performance Testing (PT)

Performance testing is conducted to evaluate the responsiveness, stability, and scalability of the home interior design application when multiple users interact with the system at the same time.

- **Tool used:** postman (runner + monitoring) is used to simulate multiple API request such as login, design generation, image upload, and fetching design data.
- **Metric observed:**
    - Average response time
    - Peak memory usage
    - Request success rate.

Table 12 performance testing

Test case ID	**PT**
Requirement ID	REQ-004
Title	AI design generate load test
Description	Generate multiple AI-based interior design at the same time
Objective	To evaluate the system performance during high AI processing request
Driver/Precondition	User login required ; postman runner configure with iteration
Test Steps	Send 10-15 POST request /generate design Monitor response time
Input	Room type , style, color
Expected Result	System generate design without delay and clash
Actual Result	All request processed successfully with stable response time
Remarks	Pass

5.5 Summary

This chapter outlined the various levels of software testing applied to the project, including unit, functional, integration, and performance testing. It demonstrated the use of Postman for API validation and provided structured guidelines for writing effective test cases.

These testing practices ensure the reliability, accuracy, and robustness of the system aligning closely with the project's objective to provide a user friendly efficient and well-performing home interior design and office design application where users can create and generate design easily.



Chapter 6 System Conversation

6. Introduction

System conversion is the method of changing from one information system to another. This involves transferring data, process and operation from an old system to a new one.

6.1 Conversion method

For this project the conversion method was adopted

1. This method involves stopping any previous or manual process and immediately starting the new system.
2. Since this project was developed from scratch and does not rely on existing or legacy system a direct implementation of the MERN stack based home interior design application was practical and cost effective.
3. Although direct conversion can be risky in real-world systems, it is suitable for this academic project as it allow quick deployment, testing and user interaction without complex migration step.

6.2 Deployment

This section presents the steps required to make the software operational.

Step followed

1. **Build process**
    - The frontend was built using npm run build command.
    - The backend was setup and tested locally using
2. **Environment configuration**
    - .env files were used to store sensitive data such as database URI and API keys.
3. **Running the application**
    - The application as executed composes up build and run all services.

6.2.1 Data conversion

The data conversion processes involve.

- **User input conversion**
  User input (room type, furniture, colors, and images) is converted into structured JSON format.
- **Image processing conversion**
  Uploaded images are processed and converted into useable format for AI based design matching.
- **Database storage conversion**
  Data is stored in structured form in the database using the MERN stack.

6.2.2 Training

To train the user, a simple user manual was prepared based on the two main use cases of the system.

Use case 1: user generating interior design

- Login as user.
- Enter room details, style, colors and furniture preference
- Click on "generate design" option
- View AI generated interior design layout

Use case 2: user uploading image for design matching

- Login as user
- Navigate to "upload image "section
- Upload room or furniture image
- View AI-images matching design suggestions

6.3 Post deployment testing

Post deployment testing included the following step.

- Tested login, registration, design generation and image upload modules through postman to ensure APIs were working as expected.
- Verified data flow and database connectivity after deployment.
- Checked all frontend routes and ensured proper navigation for user and admin roles.
- Ensured all services were running correctly with persistent data.

6.4 Challenge

Table 13 challenge

challenges	solution
Users find it difficult to design interior manually	Provide an AI-based interior design generator that suggests layout automatically
Matching furniture with room color and style is complex	Allow user to upload images for AI-based color and furniture matching
Data storage and management issues	Use structure database MySQL with proper relationship
Slow system performance with heave images	Optimize images and use efficient backend process

6.5 Summary

This chapter described the deployment of the home interior design application using a direct conversion method and Dockers for a stable environment. This chapter includes AI-based design generation, image upload and manual design features.

Post deployment testing ensured all modules worked correctly, and challenges. This system is now real world or academic use.



Chapter 7 Conclusion

7. Introduction

This chapter concludes the project by evaluating the goal defined in the beginning and highlighting future improvement. It assess whether the objective of Home Interior design and Office system have been met and suggest future enhancement

7.1 Evaluation

Below is a list of objective stated in chapter 1 and their implementation status in the final project:

Table 14 Evaluation

Objective	Status
Allow user to select to design multiple rooms in their home	Completed
Allow user to choose different design styles(e.g. modern, classic)	Completed
Provide a furniture catalog with beds, sofas, chairs, lamps	Completed
Display design preview through 3D visualization	Completed
Provide user with option to upload their room images	Completed
Enable drag and drop functionality for easy placement of furniture	Completed
Support saving and share user designs for future use	Completed

7.2 Traceability Matrix

The traceability matrix ensures that all software requirements are fulfilled by mapping them to design, code and test cases:

Table 15 Traceability Matrix

Requirement ID	**Requirement Description**	**Design Specification**	**Code**	**Test ID**
R1	User login & authentication	Login module	Login.html/ auth.js	FT 1

R2	Select room type & preferences	Room & Style Module	Room .html & style. html	FT 2
R3	AI design generate	AI design module	Generate.js	PT 1
R4	Upload images	Image upload module	Uploaded.html	IT 1
R5	Save design	Save Module	Save.js	IT 2
R6	Retrieve save design	Dashboard Module	Dashboard .html	IT 3
R7	Share design	Share Module	Share .js	FT 3

7.3 Conclusion

The Home Interior Design and Office system successfully fulfill the objective set in the early stage of development.

Table 16 Conclusion

Objectives	**Functionality Provided By System**
Allow users to select room type	Room selection e.g. bedroom, office
Provide multiple design styles	Style selection e.g. modern, classic
Generate interior design automatically	AI based design generate
Enable user to upload room images	Image upload functionality
Allow user to saved design	Save design
Provide access to save design	Save design can retrieve
Allow user to share design	Share functionality provided

7.4 Future Work

Here are some potential areas for future enhancements:

- Develop a furniture marketplace feature for purchasing items used in design
- Create a mobile app application to improve accessibility and user reach
- Provide multi-language support for a wider audience
- Add more design template and customization option for user
- Improve system performance to reduce response time and increase efficiency
- Enhance system scalability to support a large number of concurrent users

Appendix A

Use Case Description Template

Appendix A Use Case Description

Use Case Template: AI interior design

Table 17 Use Case 1

Use Case ID:	UC-1
Use Case Name:	Generated AI interior design
Actor:	User
Description:	This use case describes how users generate an interior design automatically using AI by providing room details or uploading room image. This system analyzes the input and produces a fully furnished room design.
Trigger:	User click on Generate Design button
Preconditions:	1. User must be logged into system 2. System must be connected to AI processing module 3. User must provide either room detail or upload an image
Postconditions:	1. Interior design successfully generated 2. Designed layout is displayed to the user 3. User can save or share the design
Normal Flow:	1. User log into the system 2. User Select room type (e.g. bedroom, kitchen) 3. User Select design preferences (e.g. style, color) 4. User optionally upload room images 5. User click generate design 6. System send input to AI module 7. AI processes the input & generate design 8. System display and preview design in 3D 9. User can choose save or share design
Alternative Flows:	**A1:** Missing input Step 3 or 4 user does not provide require input → Show error message prompt correction and upload image **A2:** AI processing failure Step 7 if AI fails to generate design → System display error message and ask to retry **A3:** invalid image upload Step 4 if unsupported file format

	→ Invalid image format
Business Rules:	1. Only registerd user can generate design 2. Input value (room type, style, color) 3. Image size and format must meet system requirement 4. AI processing must complete within acceptable time
Assumptions:	1. Internet connection is available 2. AI module is properly configured and running

Use Case Template: Design interior room

Table 18 Use Case 2

Use Case ID:	UC-2
Use Case Name:	Design interior room
Actor:	User
Description:	This use case describes how users design a interior room by selecting design type, defining room space, adding furniture and visualize the design in 3D.
Trigger:	User login and start the interior design process
Preconditions:	1. User must be registered and logged in 2. System must be available and functional
Postconditions:	1. Interior room design is created 2. Design is visualized in 3D 3. User can save the design
Normal Flow:	1. User register into the system 2. User select interior design type 3. User Select area room type 4. User define room space 5. User add furniture to the room 6. User visualize the design in 3D 7. User save the final design 8. User can share the design

Alternative Flows:	**A1:** User not registered Step 1 system ask user to register first **A2:** invalid room data Step 4 if invalid dimension entered → system show error and ask for correction **A3:** Save Failure Step 7 if saving fails → system display error and ask to retry
Business Rules:	1. Only logged in user can design and save room 2. Room dimension must be valid 3. Furniture selection must match room size constraint
Assumptions:	1. Database is properly connected 2. Internet connection is available 3. System support 3D visualization

Appendix B

Coding standards

Appendix B General Coding standards & Guideline

The following coding standard and guideline were followed during the development of Home interior design & Office System to ensure code quality, readability and maintainability.

1. Naming Convention

- A consistent naming convention is followed throughout the project.
- Common styles include:
    - Snake Case: variable one
    - Pascal Case: VariableOne
    - Camel Case: variableOne
    - Hungarian Notation: arrFurnitureList

2. Scope Based Naming

- Variable name reflect their scope and purpose (e.g., privateField, constValue)

3. Immutability

- Read-only variable are used where values should not change after initialization

4. Controlled Access

- Getter method are used for properties that should not be modified externally

5. Function Naming

- Function named are clearly based on their functionality (e.g., generateDesign(), uploadImage())

6. Comments and Documentation

- Meaning ful comments are added to explain complex logic
- Example:

```
/* This function generate an interior design based on user input */
Function generateDesign() {
}
```

- Avoid unnecessary comments for obvious code

7. Code Indentation

- Proper indentation is maintained to improve readability and structure

8. File and Folder Naming

- Consistent naming is used for file and folder
- File name reflect their content(e.g. login.js, desigModule.js)

9. Class Structure

- Classes are properly structured with clear separation of properties and method

10. Code Organization

- Code is logically divided into module such as:
    - UI Layer
    - Business Logic Layer
    - Data Layer
- Each class/interface is placed in a separate file with matching names

11. Scope Management

- Variable and function are defined within the smallest possible scope

12. Access Modifiers

- Appropriate Access Modifiers(public, private, protected) are used where neccessary

13. High Cohesion & low coupling

- Each module perform a specific task(High Cohesion)
- Dependencies between module are minimized(low coupling)

14. DRY Principal (Don’t repeat yourself)

- Code duplication is avoided
- Reusable function and component are created
- Methods are keep short and focused

15. Single Responsibility Principal

- Each function or class perform only one responsibility

16. Extensibility

- Code is written in a way that allow easy extension without modifying existing functionality

17. Exception Handling

- Error and exception are properly handeld to avoid system craches

18. Logging

- Important events and error are logged for debugging and monitoring

19. Consistent logging standards

- A uniform logging format is followed across the application

Appendix C

Prototype

Appendix C: Application Prototype

1. Prototype overview

The application prototype provides a visual representation of the system's layout and behavior. It covers the following core modules:

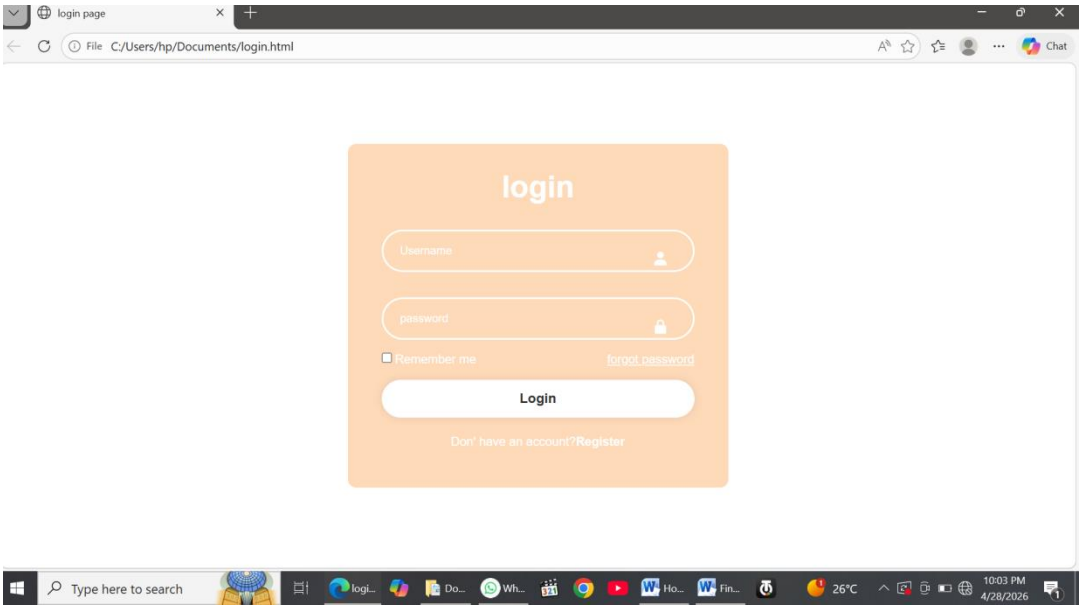
- User Select Area
- User Select Design
- User can preview design in 3D
- User automatically create design in AI
- User can save and share design

2. Tools Used for Prototype

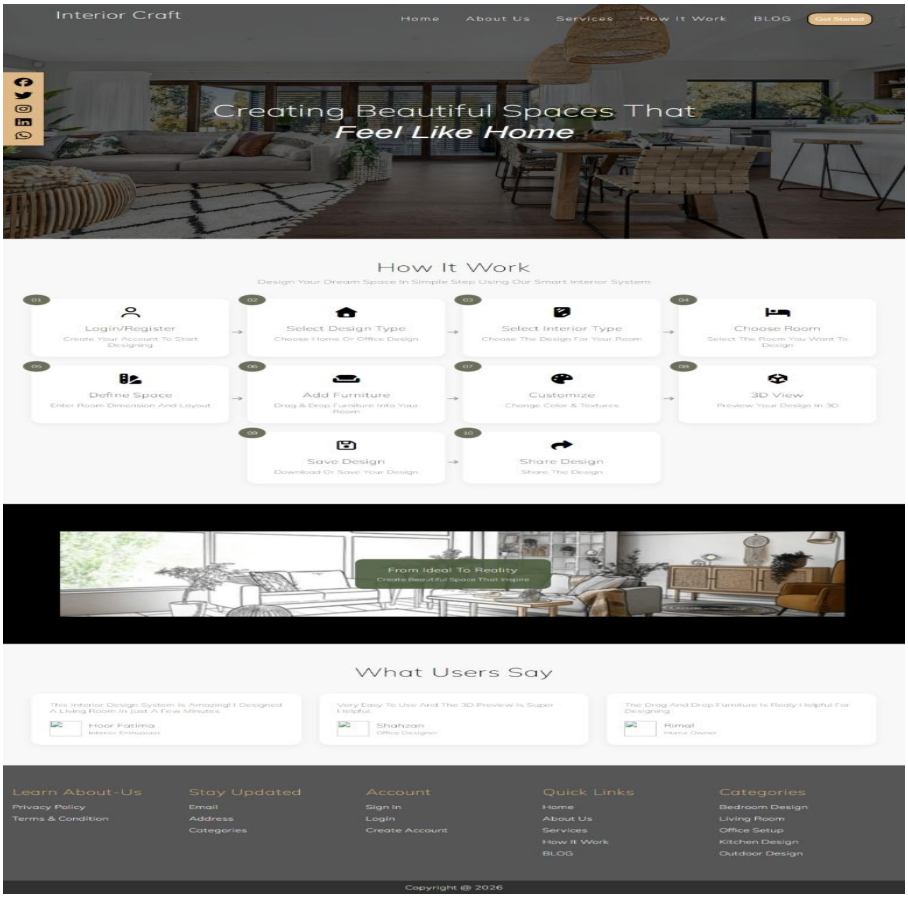
- Frontend: HTML, CSS, JS
- Backend: PHP
- Database: MySQL
- Version Control: Github

3. Screenshot of key interfaces

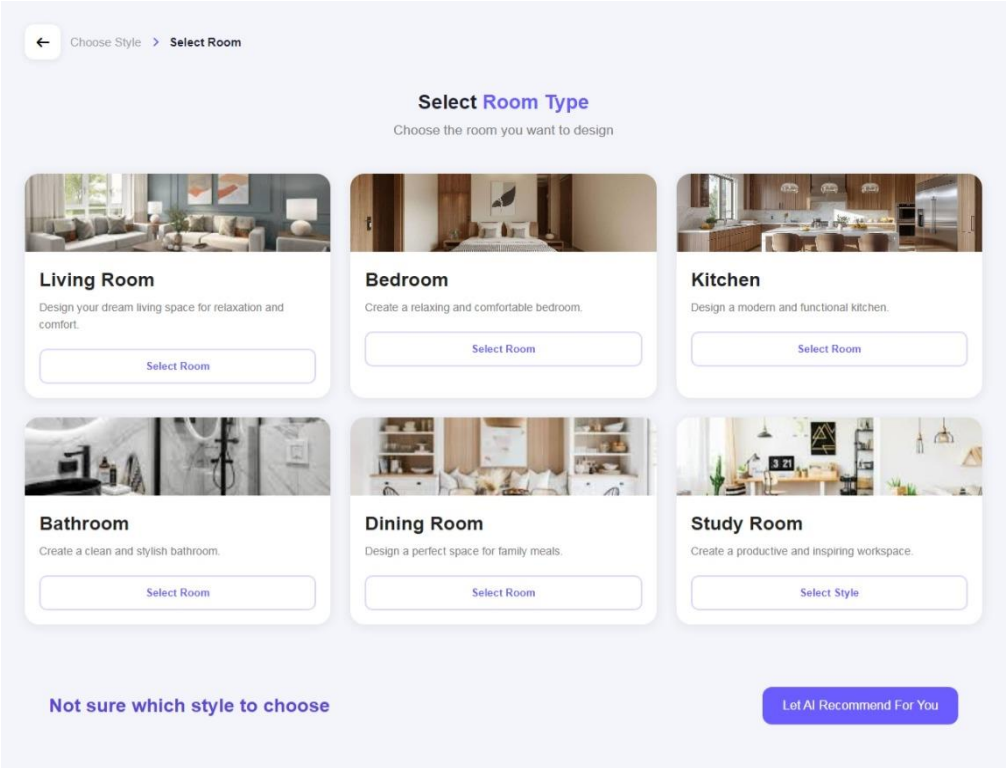
1. User Login Page



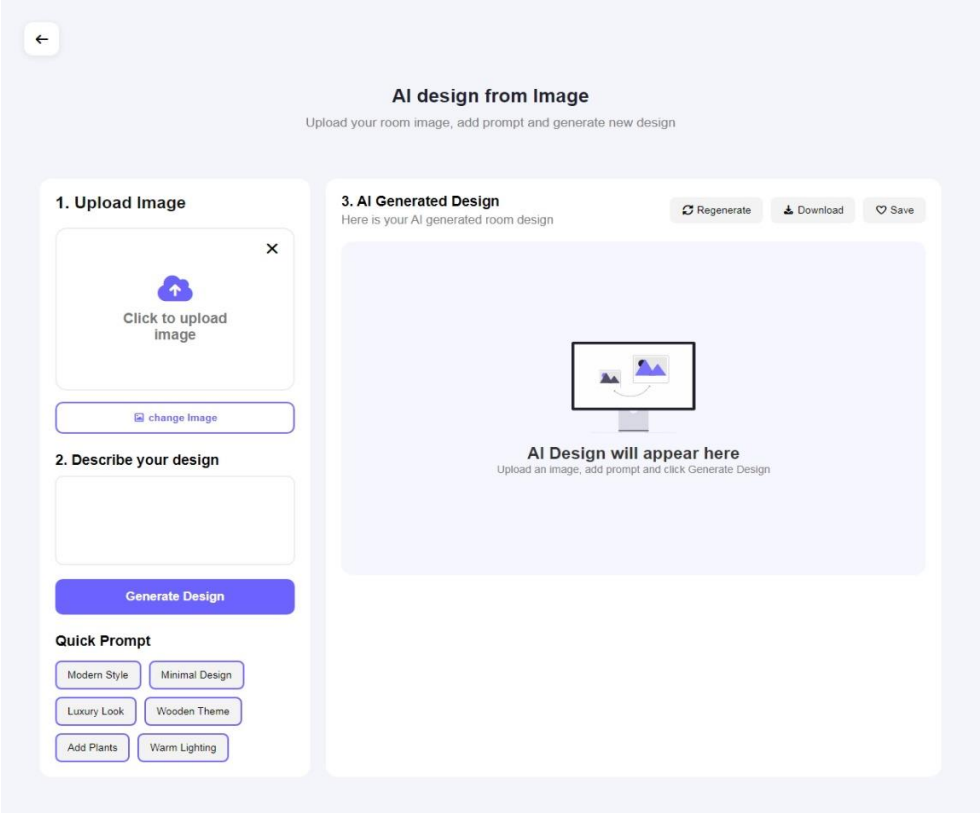
2. Home Page



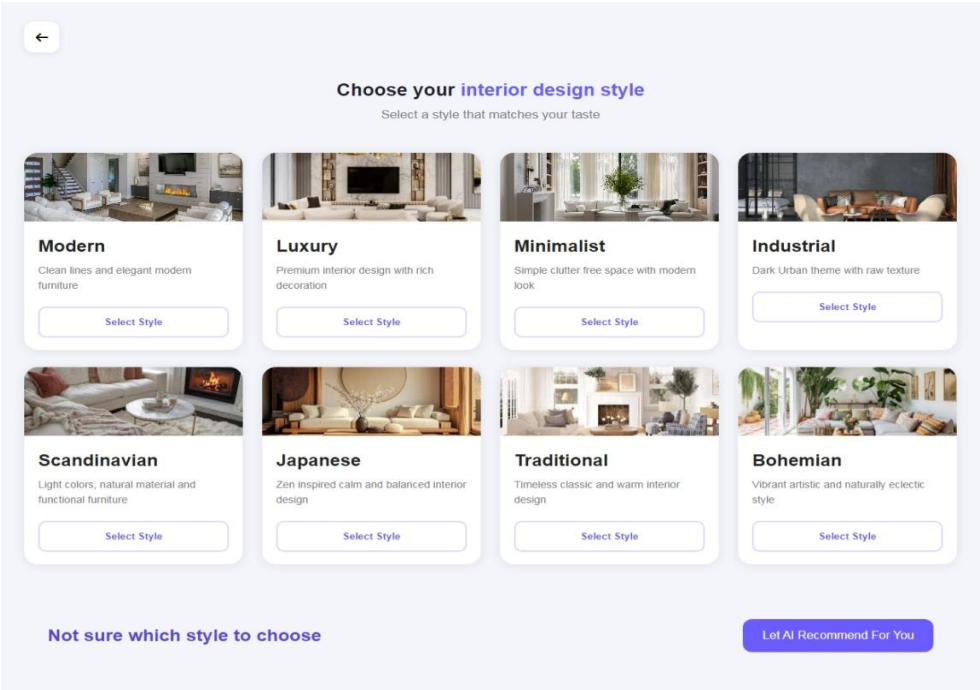
3. Selection Area



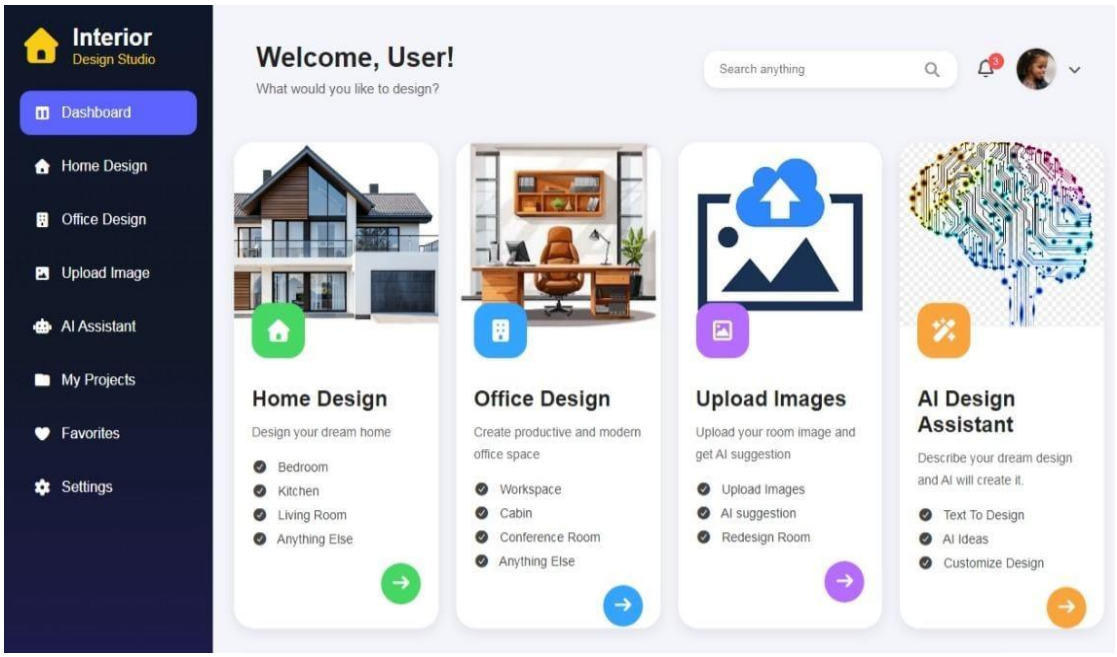
Upload a photo



4. Select Design



5. Create Design



4. Navigation Flow

Here's the typical Flow of action:

Home Page → Login Page → Select Design → Select Area → Create Design → Save Design → Share Design

5. Responsive Design

This Application is fully responsive and optimize for:

Desktop

Tablet

Mobile App